



US 20250286270A1

(19) **United States**

(12) **Patent Application Publication**
Hintermann et al.

(10) **Pub. No.: US 2025/0286270 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **EYEWEAR WITH SLOT-RING ANTENNA**

G02C 11/00 (2006.01)

H01Q 13/10 (2006.01)

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(52) **U.S. Cl.**

CPC **H01Q 1/273** (2013.01); **G02B 27/0176**
(2013.01); **G02C 11/10** (2013.01); **H01Q**
13/10 (2013.01); **G02B 2027/0178** (2013.01)

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(21) Appl. No.: **19/217,854**

(22) Filed: **May 23, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/583,678, filed on
Feb. 21, 2024, now Pat. No. 12,347,926, which is a
continuation of application No. 17/386,716, filed on
Jul. 28, 2021, now Pat. No. 11,929,548.

Publication Classification

(51) **Int. Cl.**

H01Q 1/27 (2006.01)

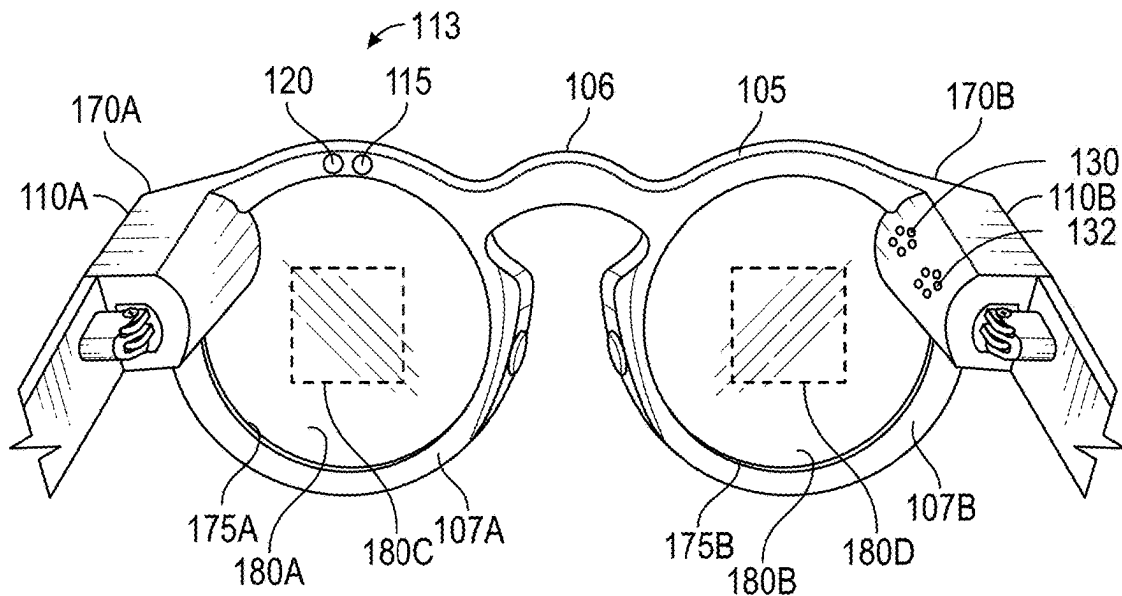
G02B 27/01 (2006.01)

(57)

ABSTRACT

Eyewear with a slot ring antenna generating orthogonal E-fields. The slot ring antenna has a first portion including an inner ring and an outer ring generating a first direction E-field, and a second portion having a respective inner ring and the outer ring generating a second direction E-field. The outer ring encompasses the respective inner rings, where a first slot is defined between the respective inner ring and the outer ring. A second slot is encompassed by the respective inner ring, which may be a cutout. The slot ring antenna encompasses eyewear optical assemblies. In one example, the first portion encompasses a first optical assembly, and the second portion encompasses a second optical assembly. A common ground plane of the outer ring may extend proximate a bridge of the eyewear.

100 →



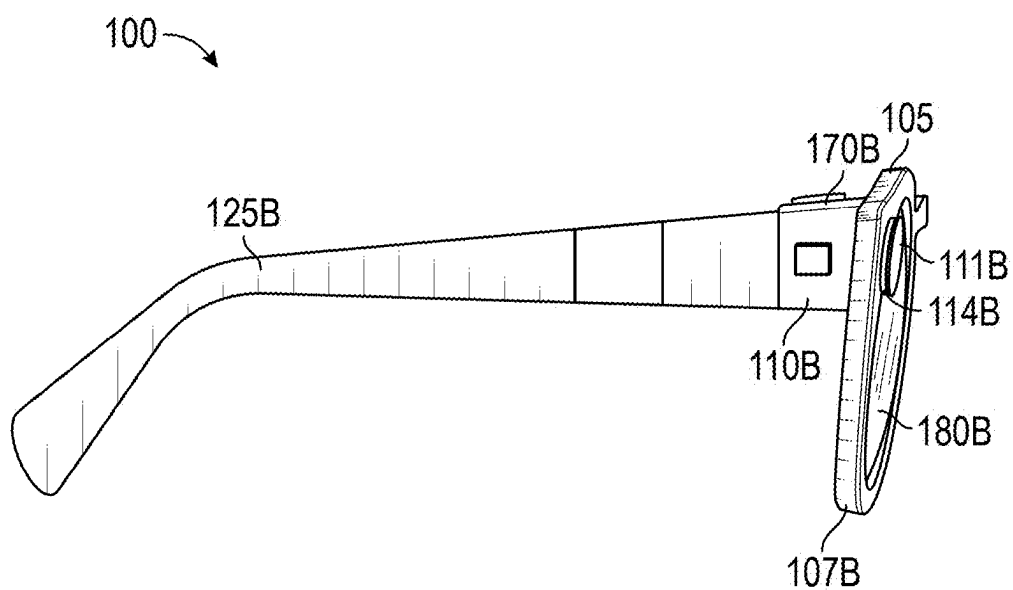
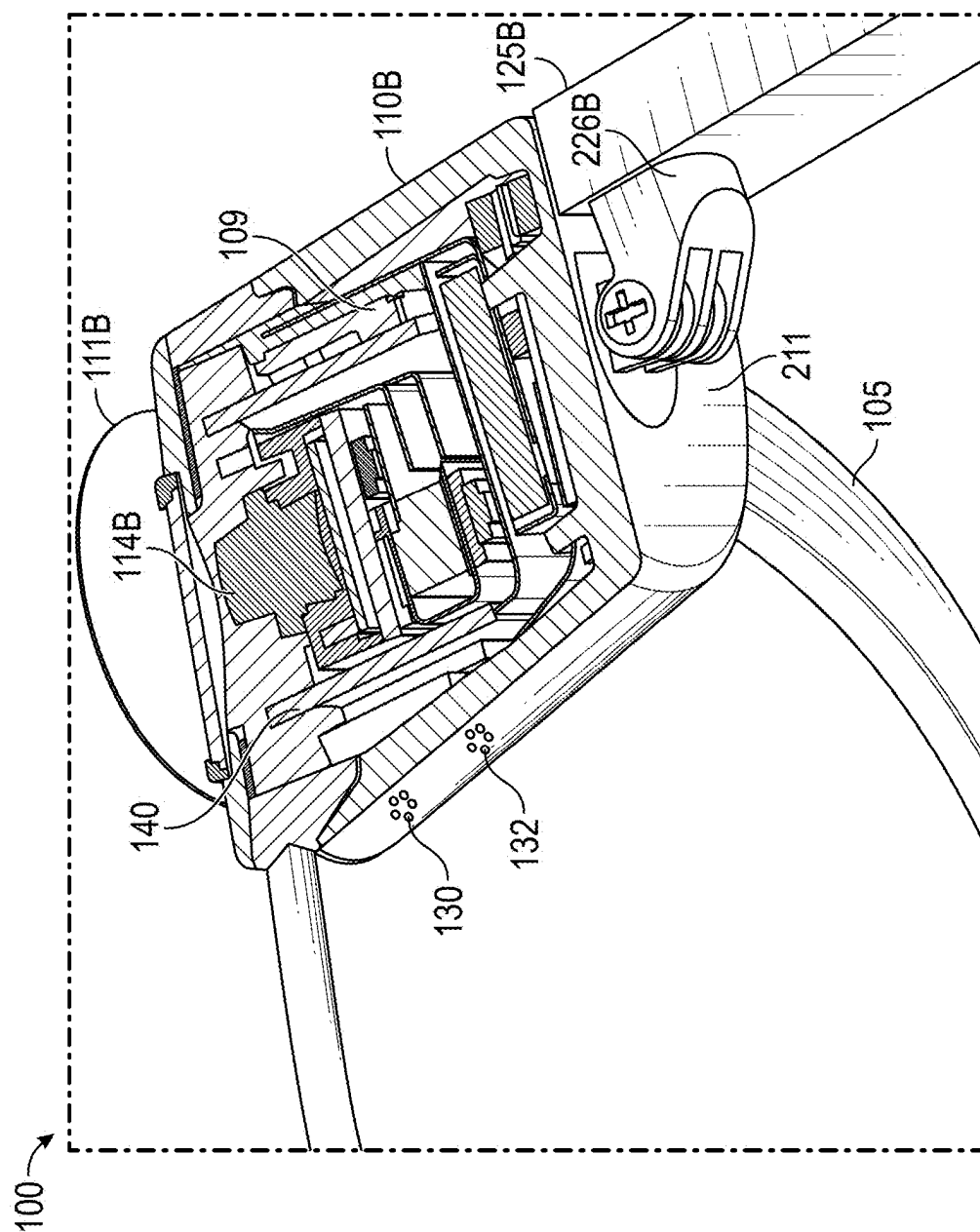


FIG. 1A

**FIG. 1B**

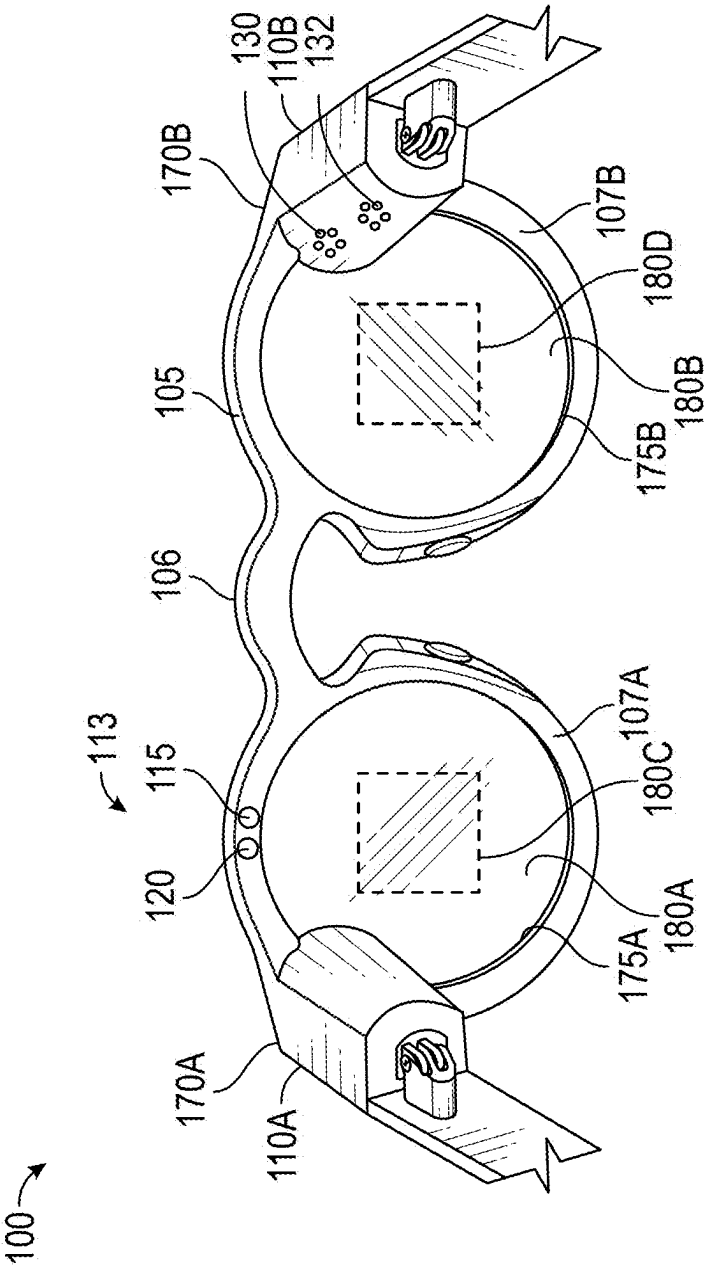


FIG. 2A

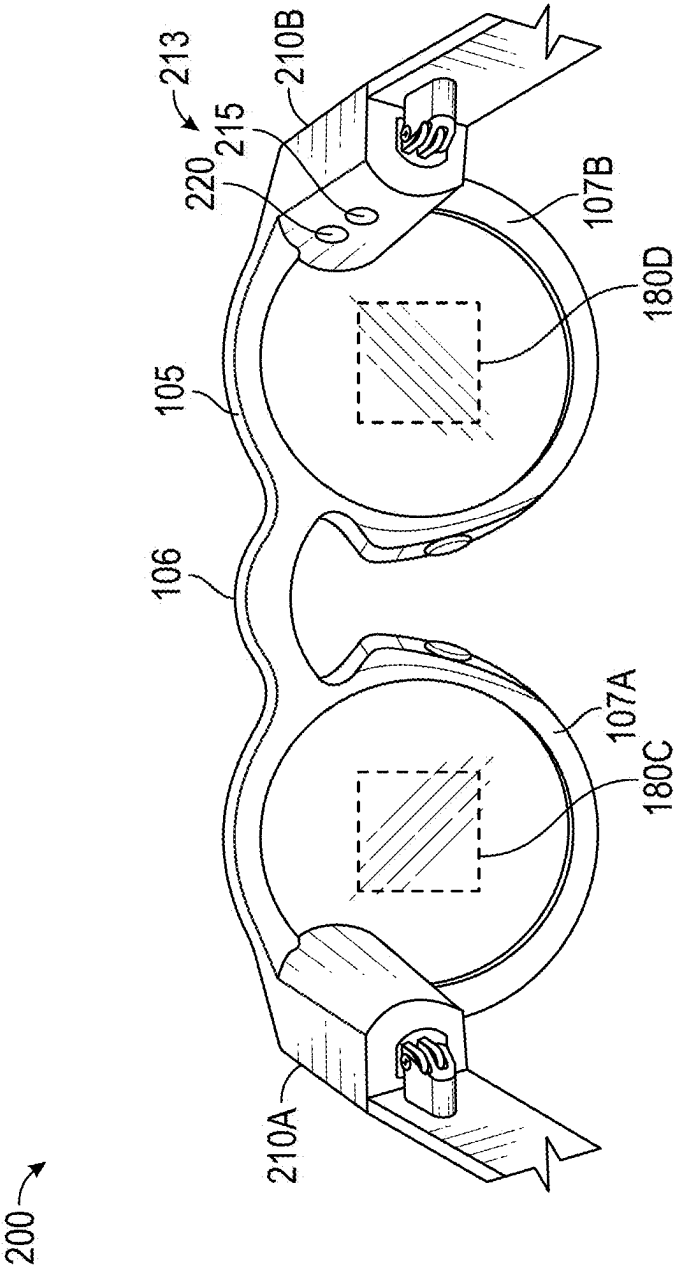


FIG. 2B

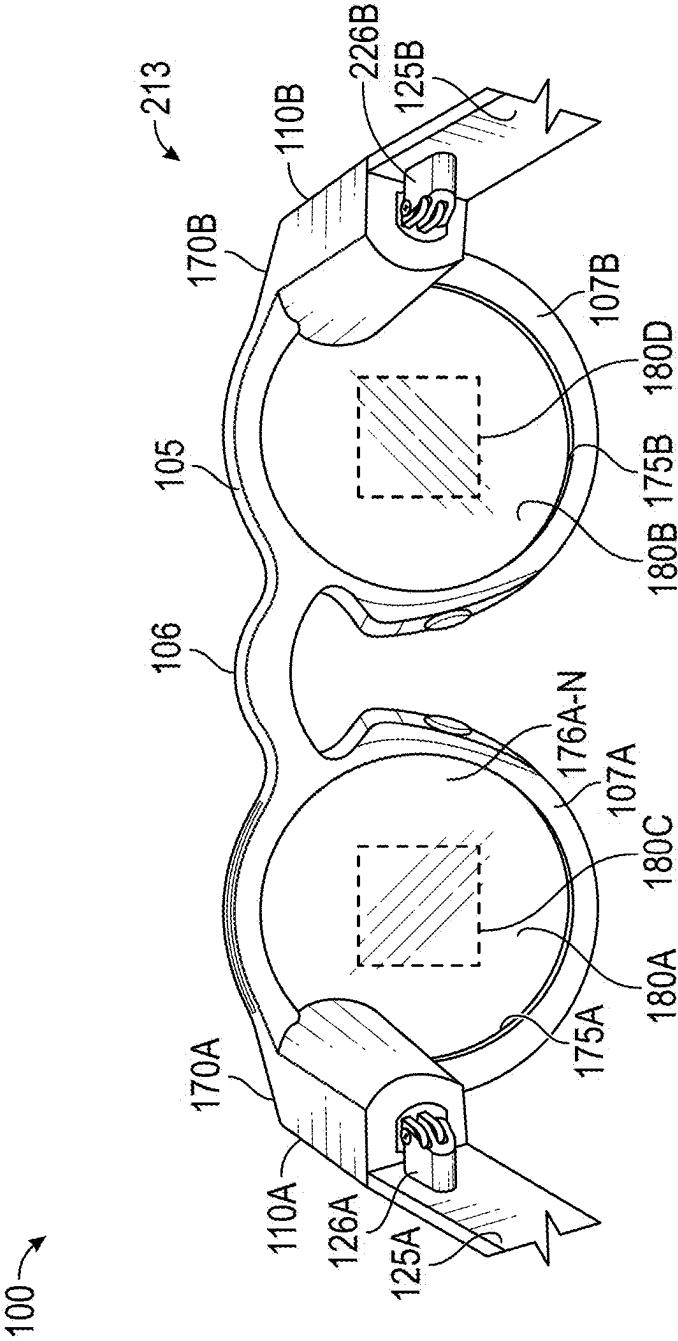


FIG. 2C

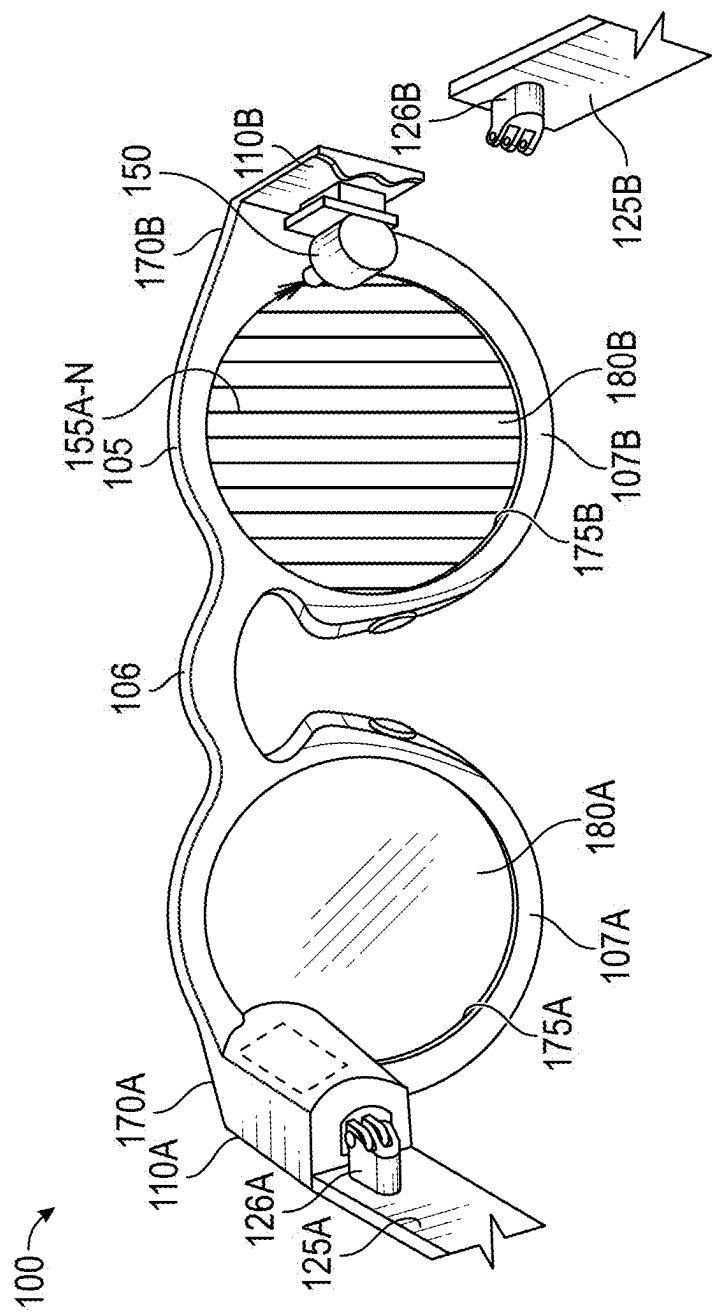


FIG. 2D

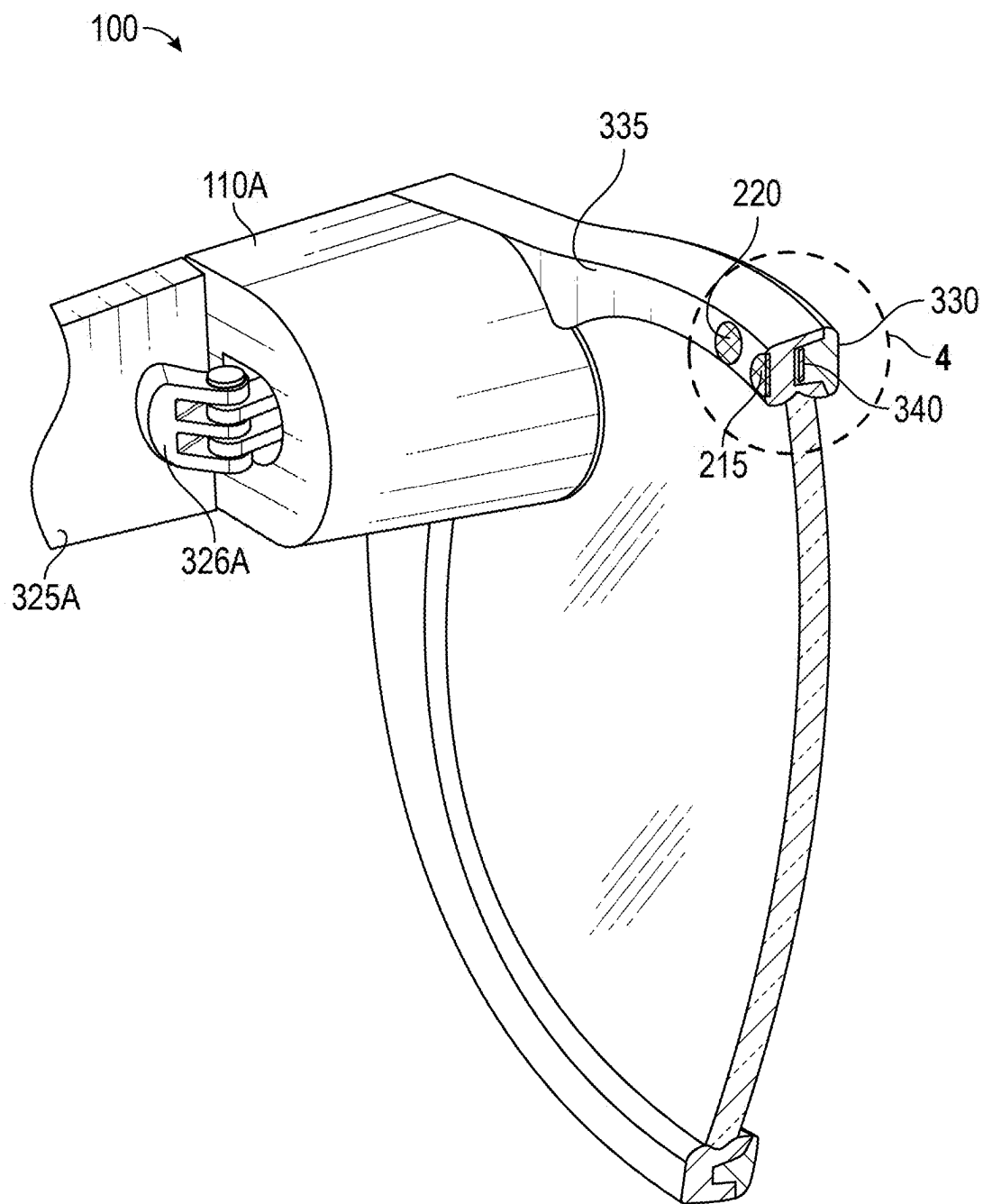


FIG. 3

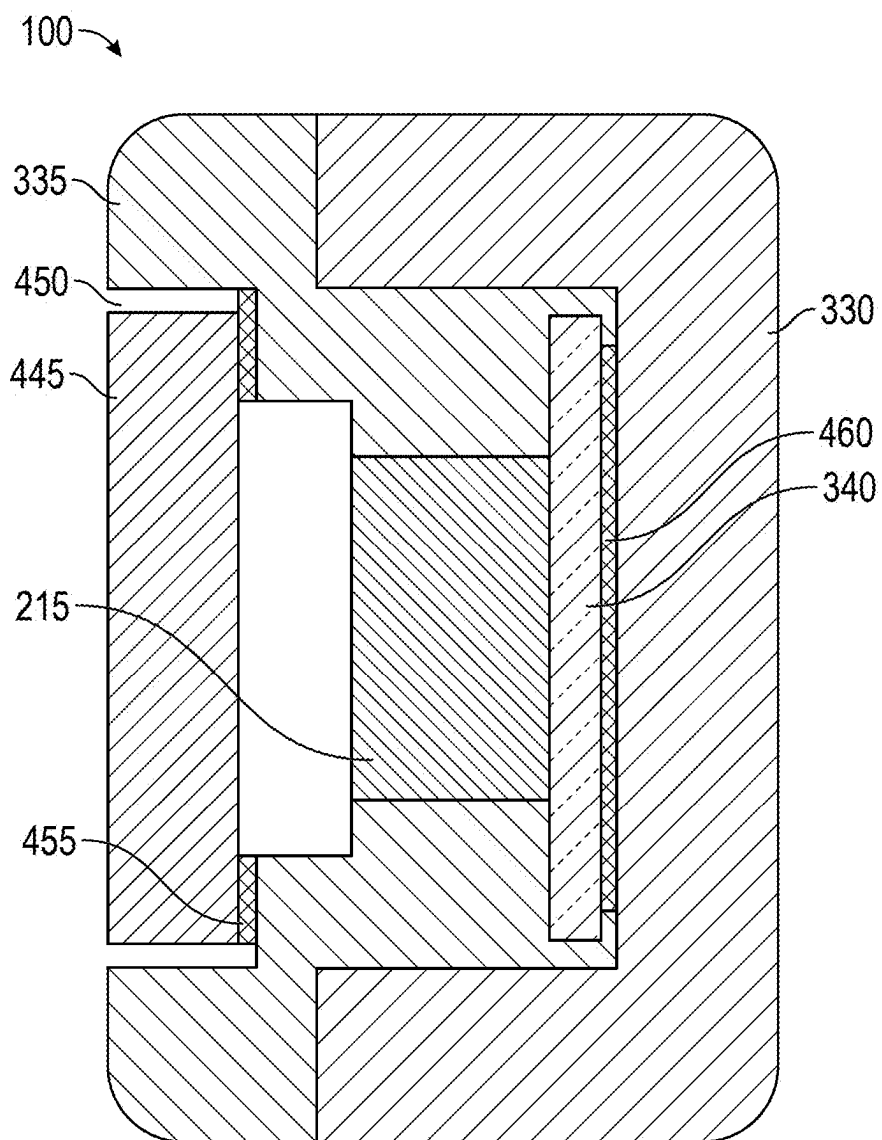


FIG. 4

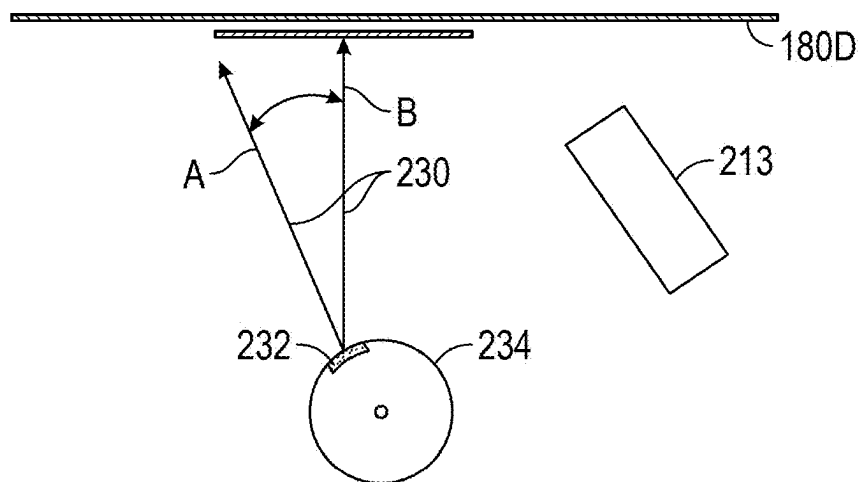


FIG. 5

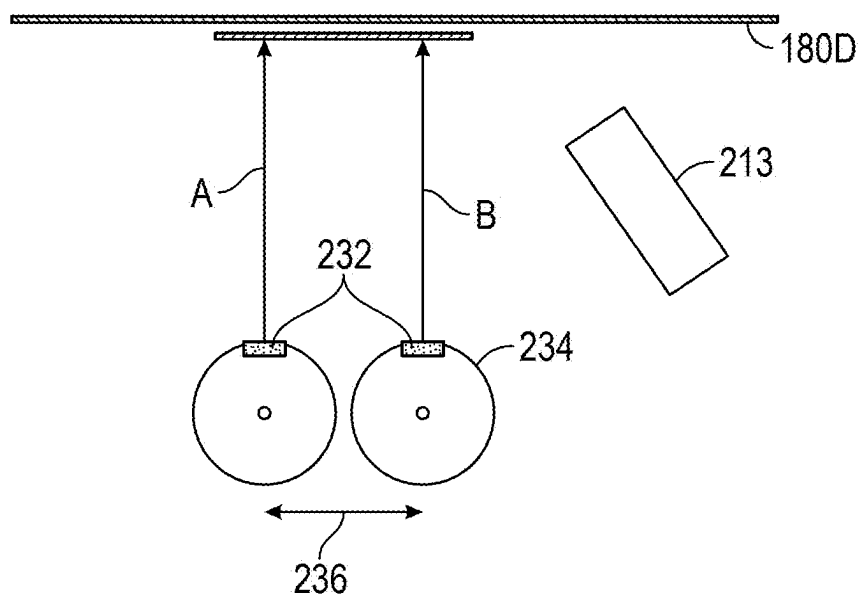


FIG. 6

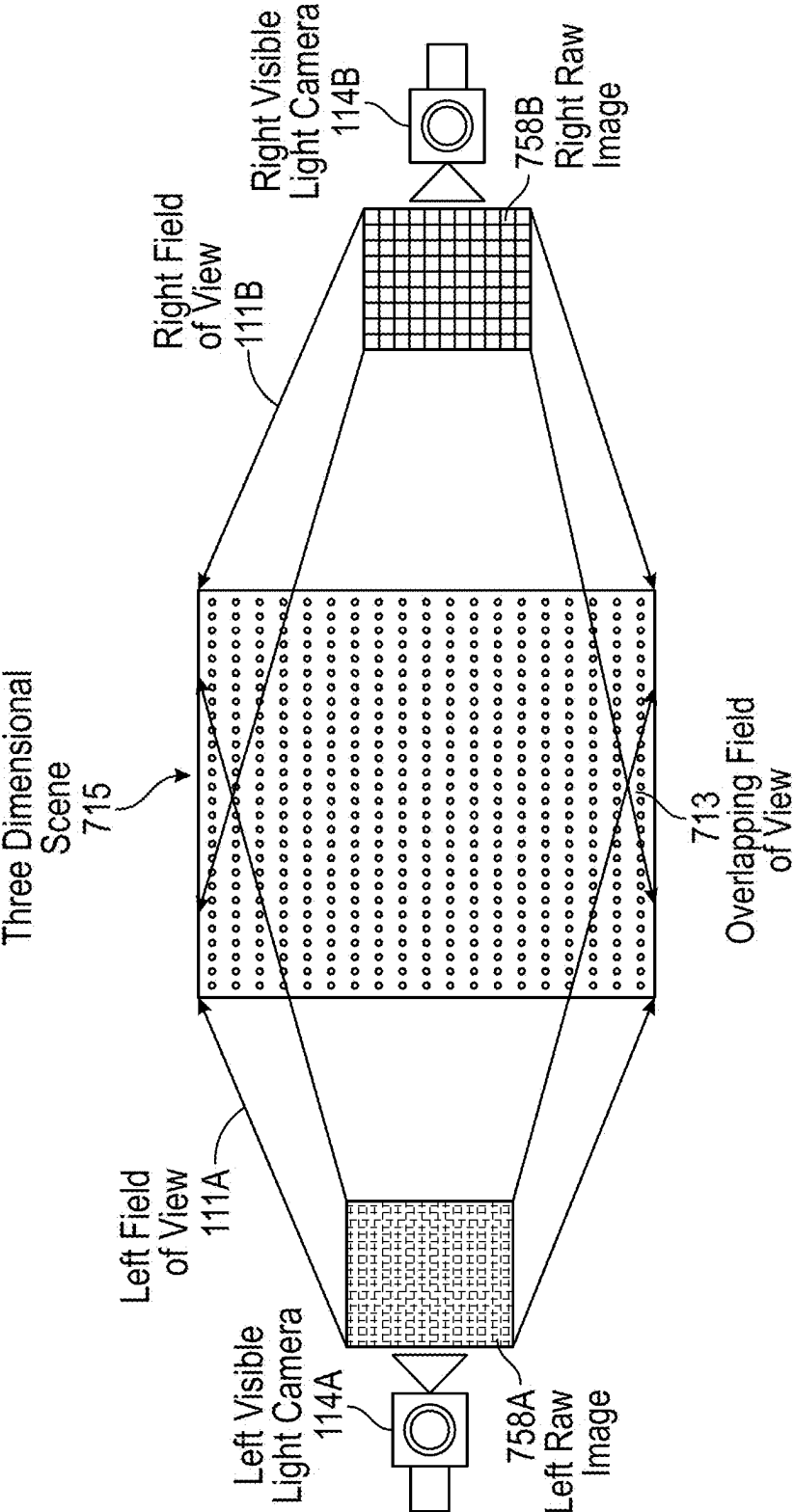


FIG. 7

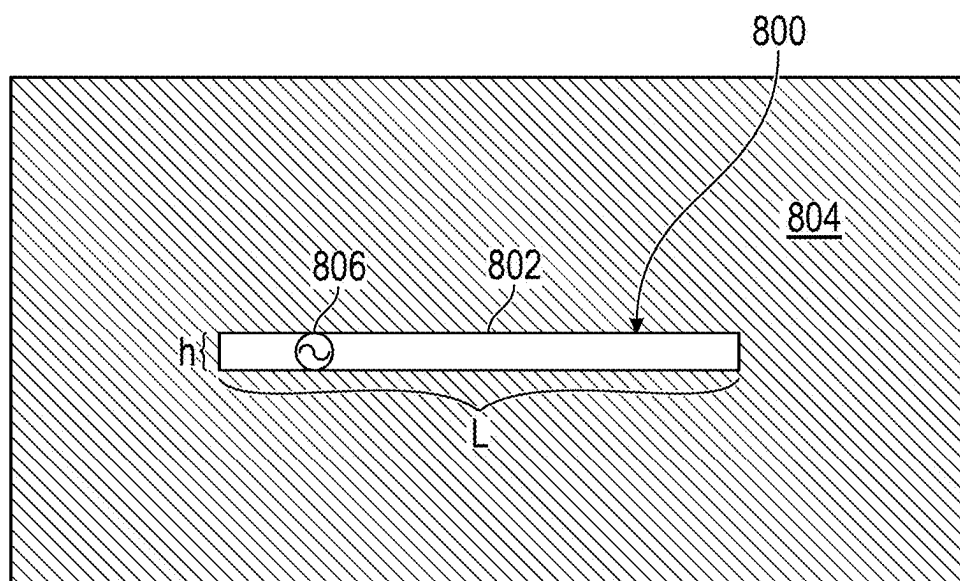


FIG. 8A

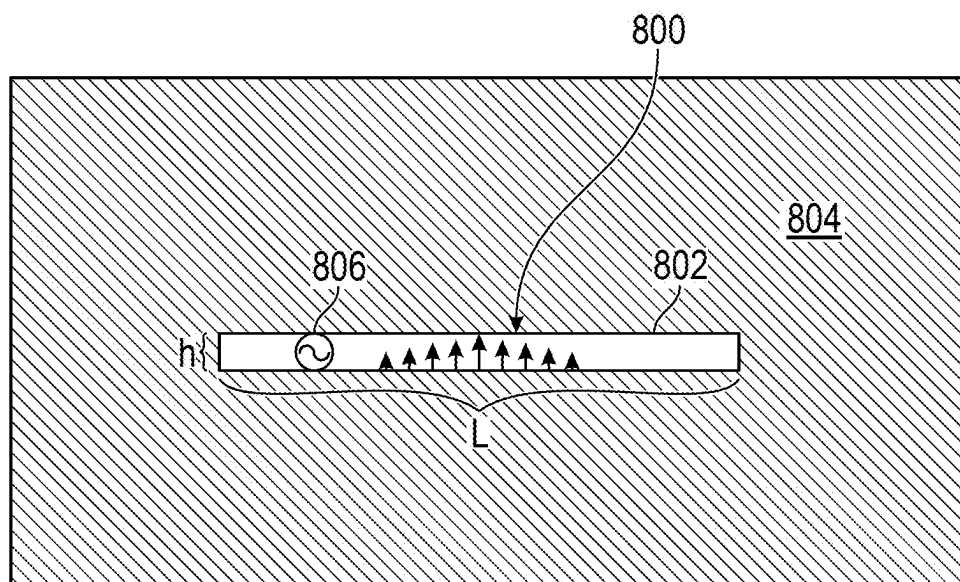


FIG. 8B

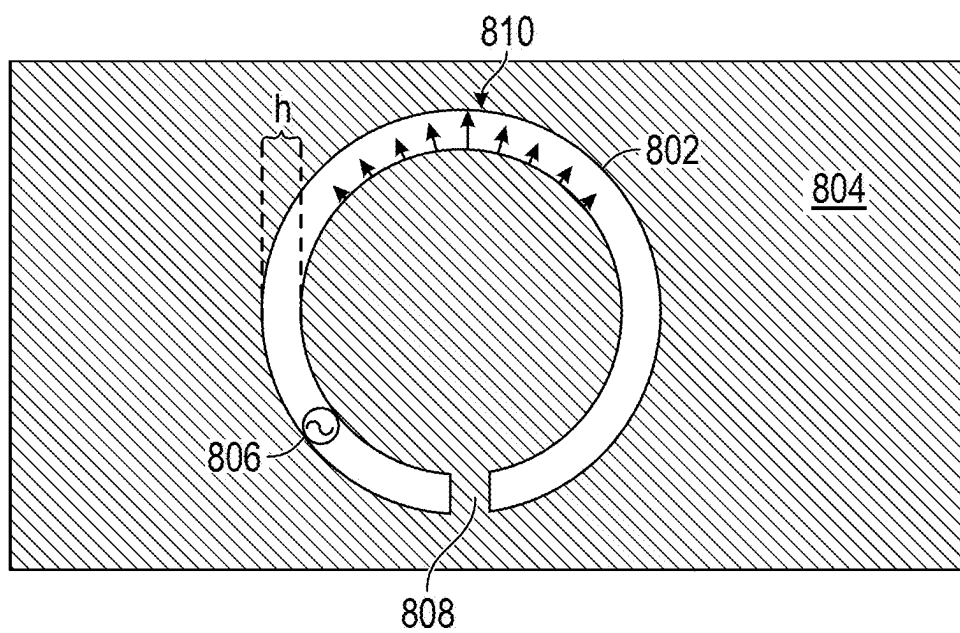


FIG. 8C

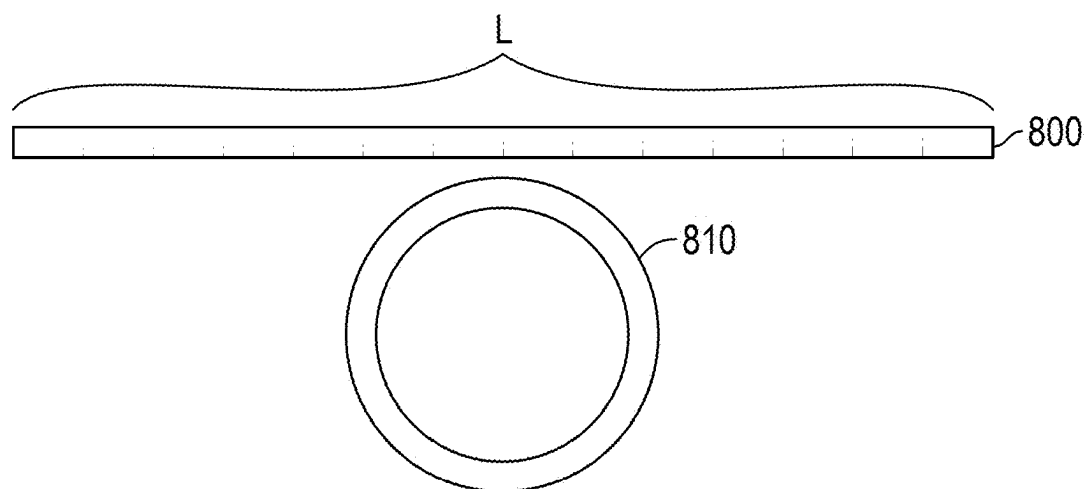


FIG. 8D

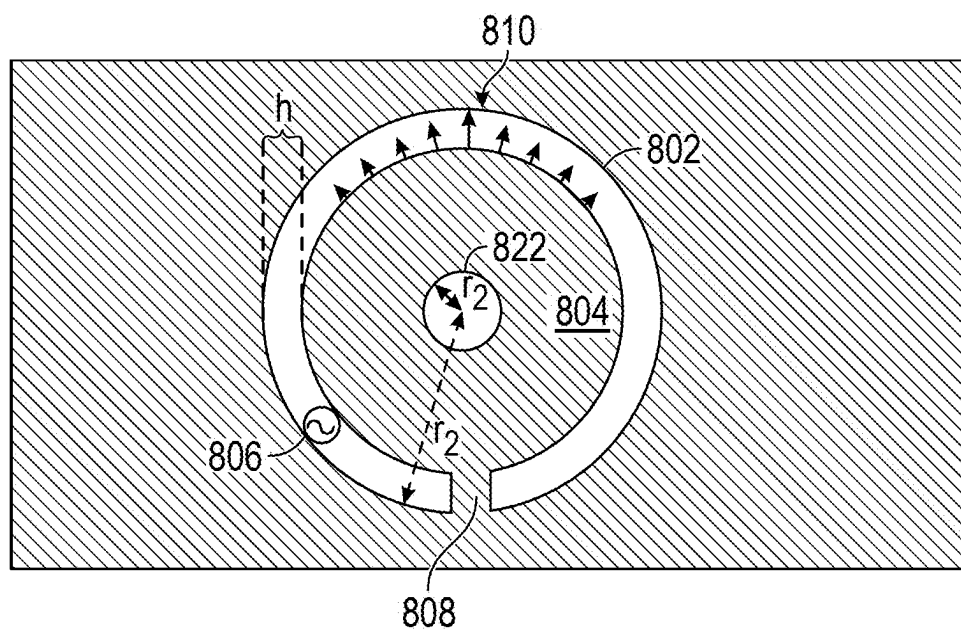


FIG. 8E

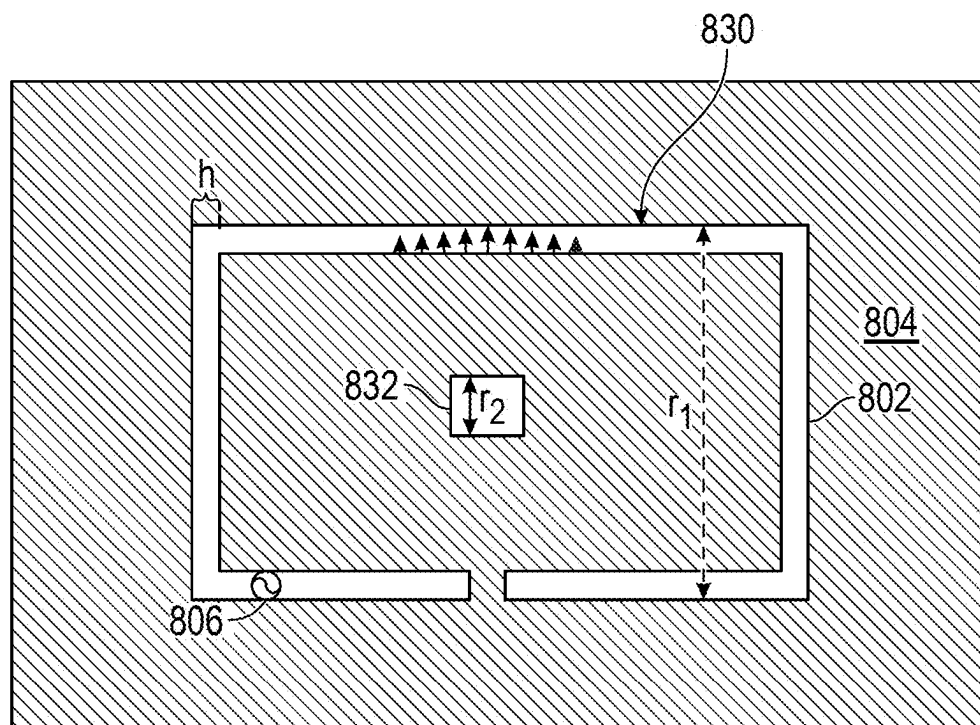


FIG. 8F

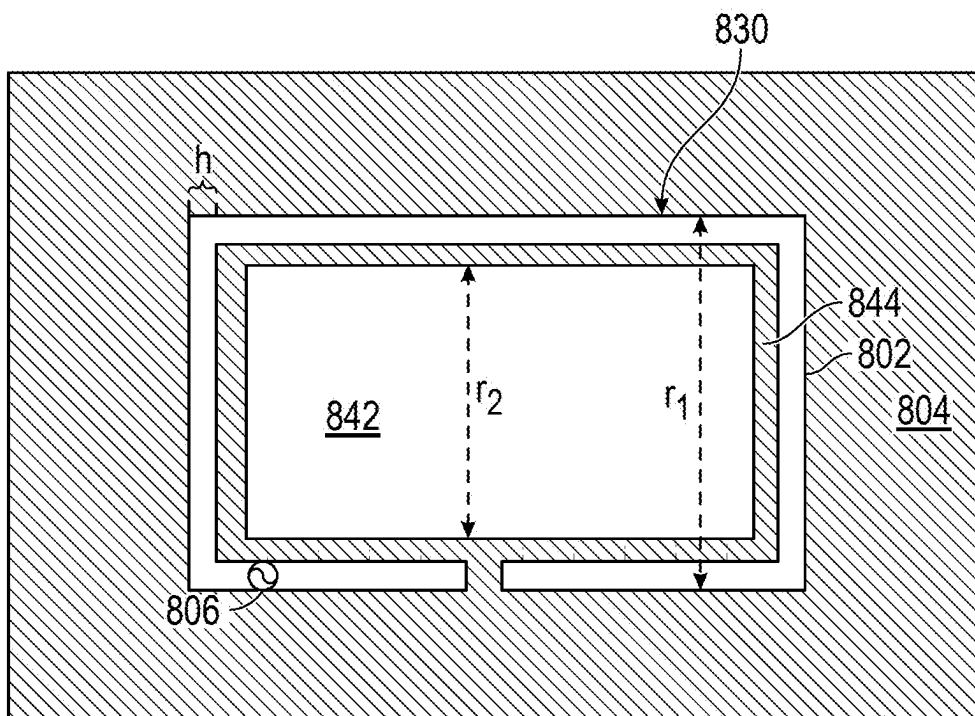


FIG. 8G

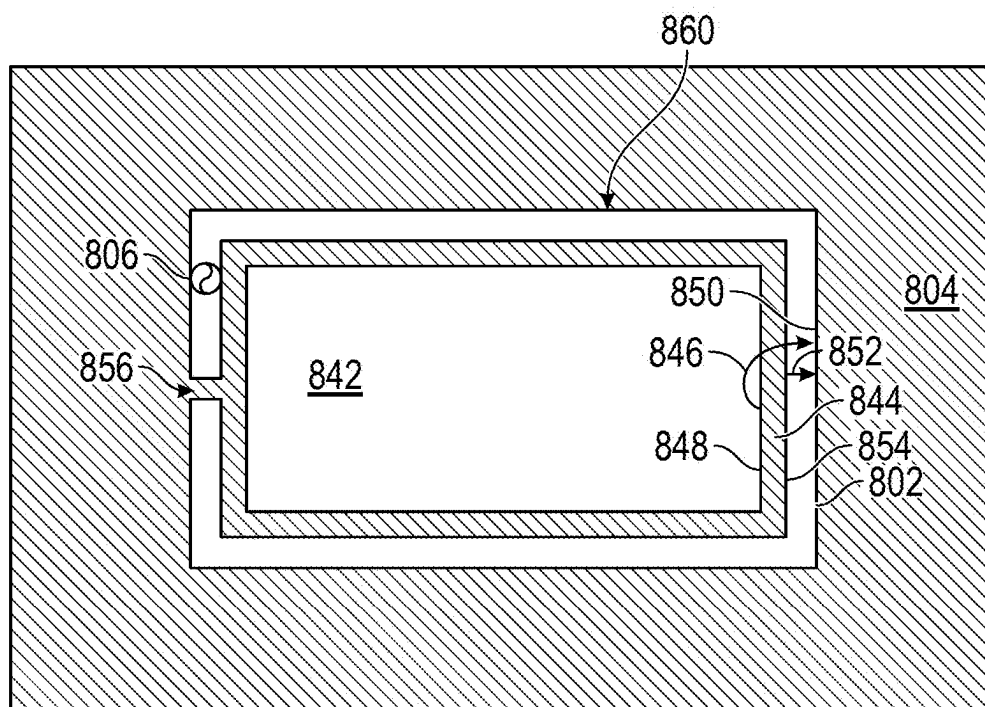


FIG. 8H

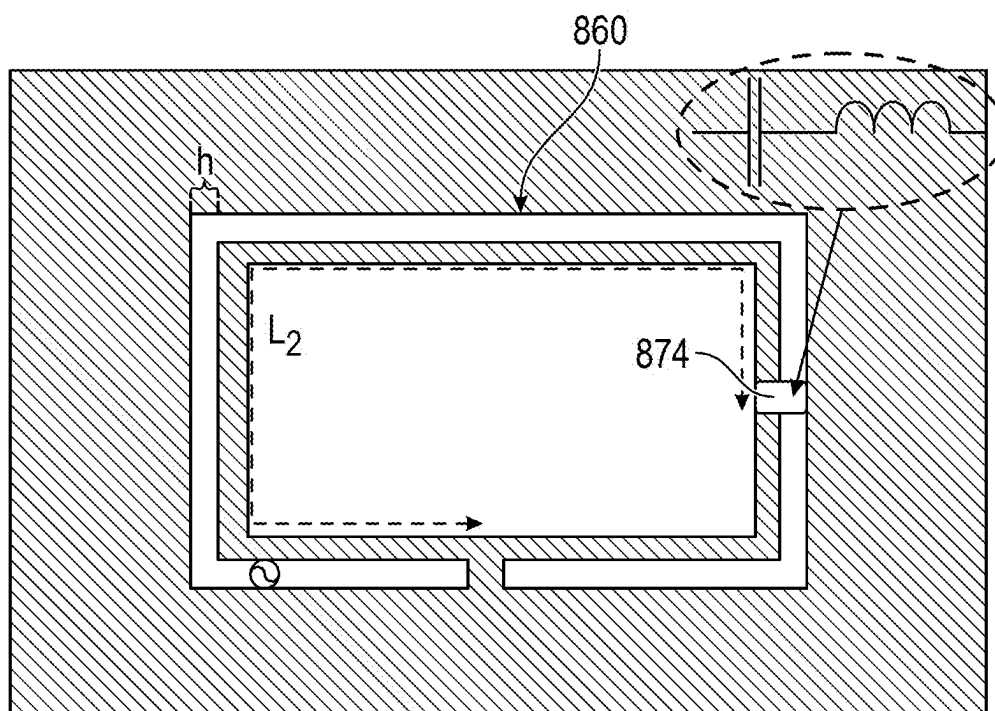


FIG. 8I

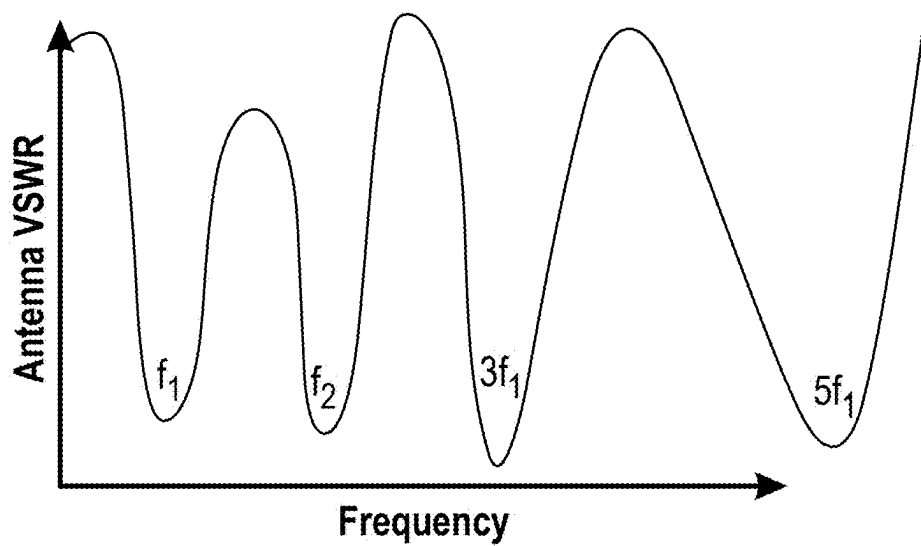


FIG. 8J

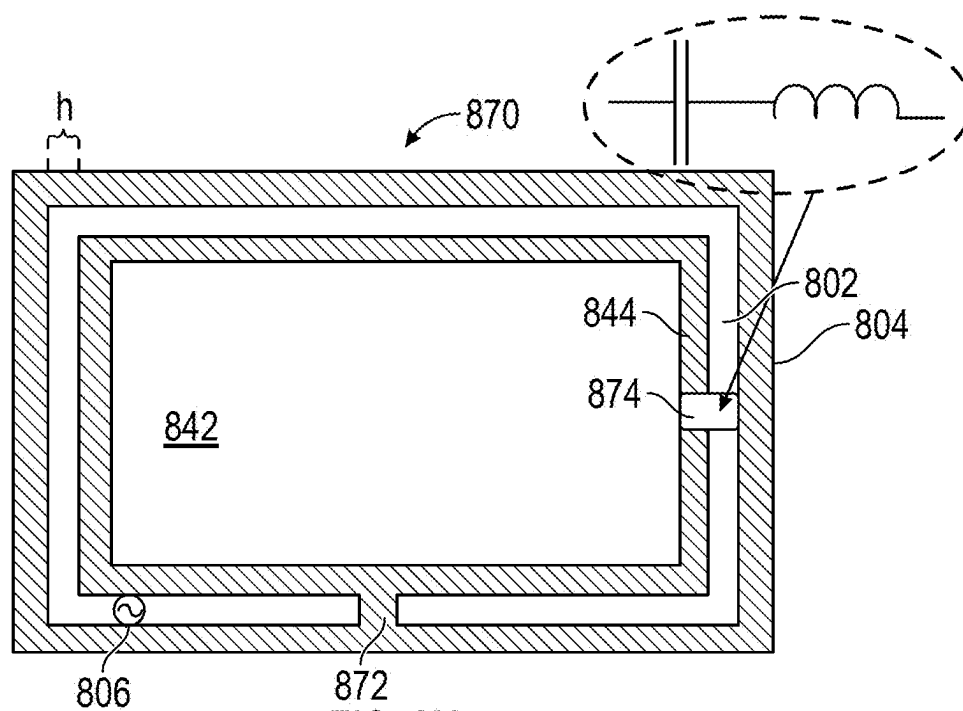
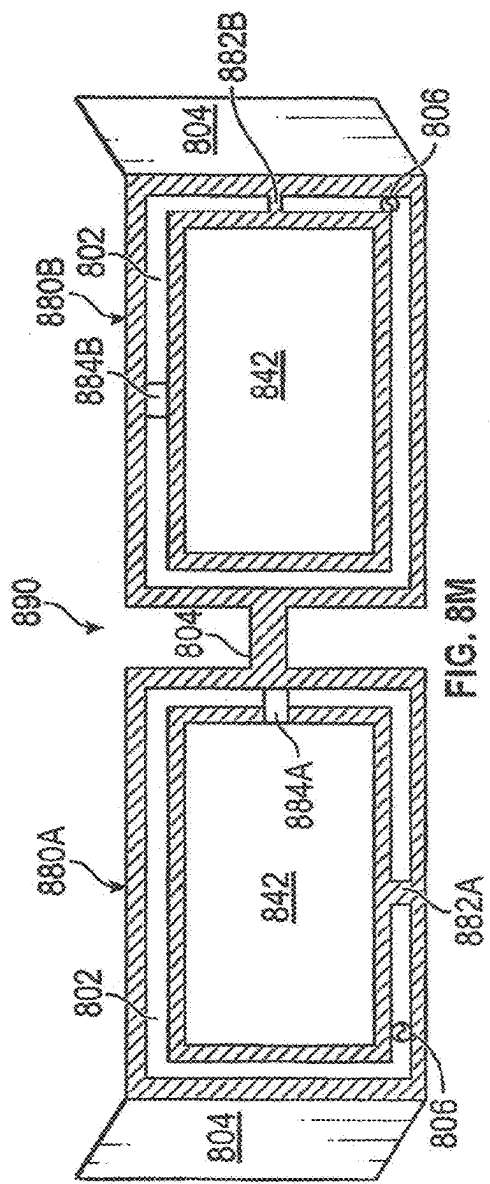
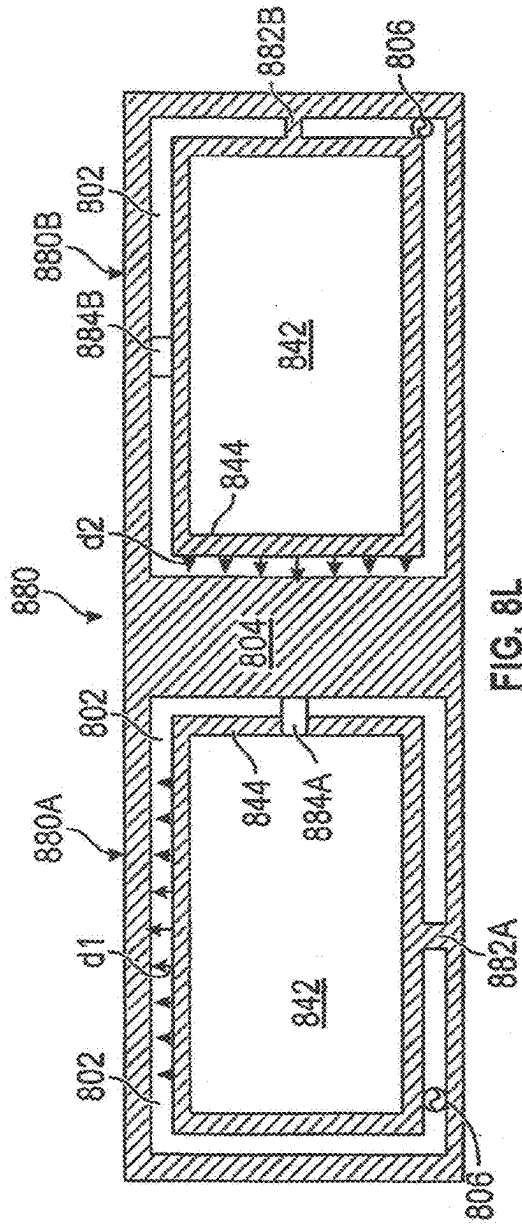


FIG. 8K



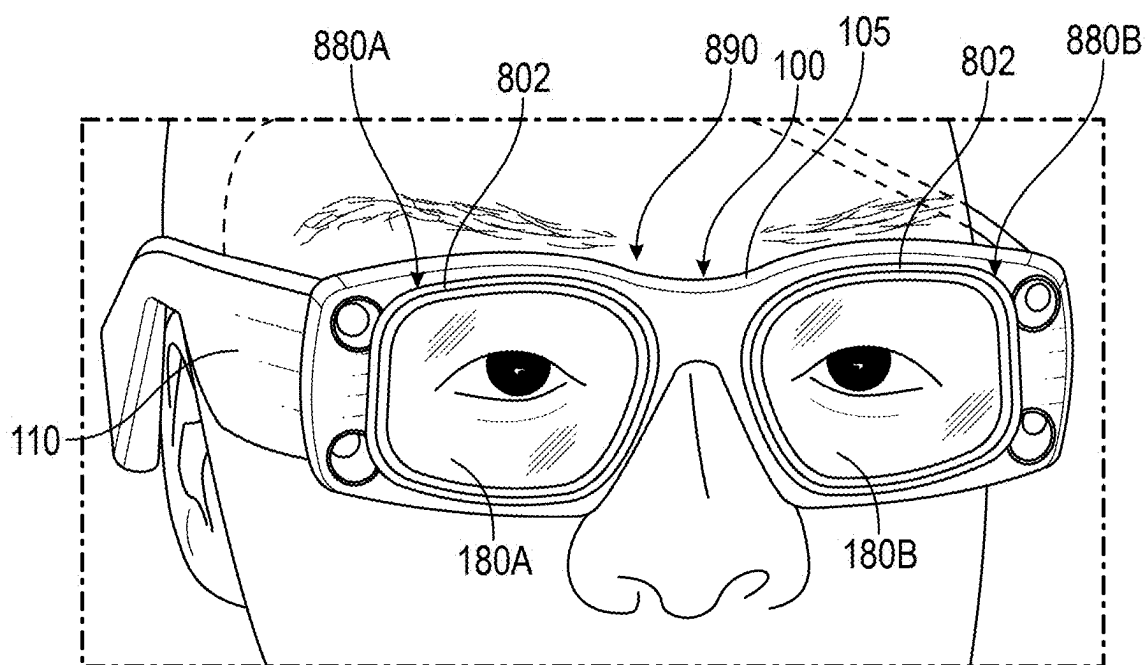


FIG. 8N

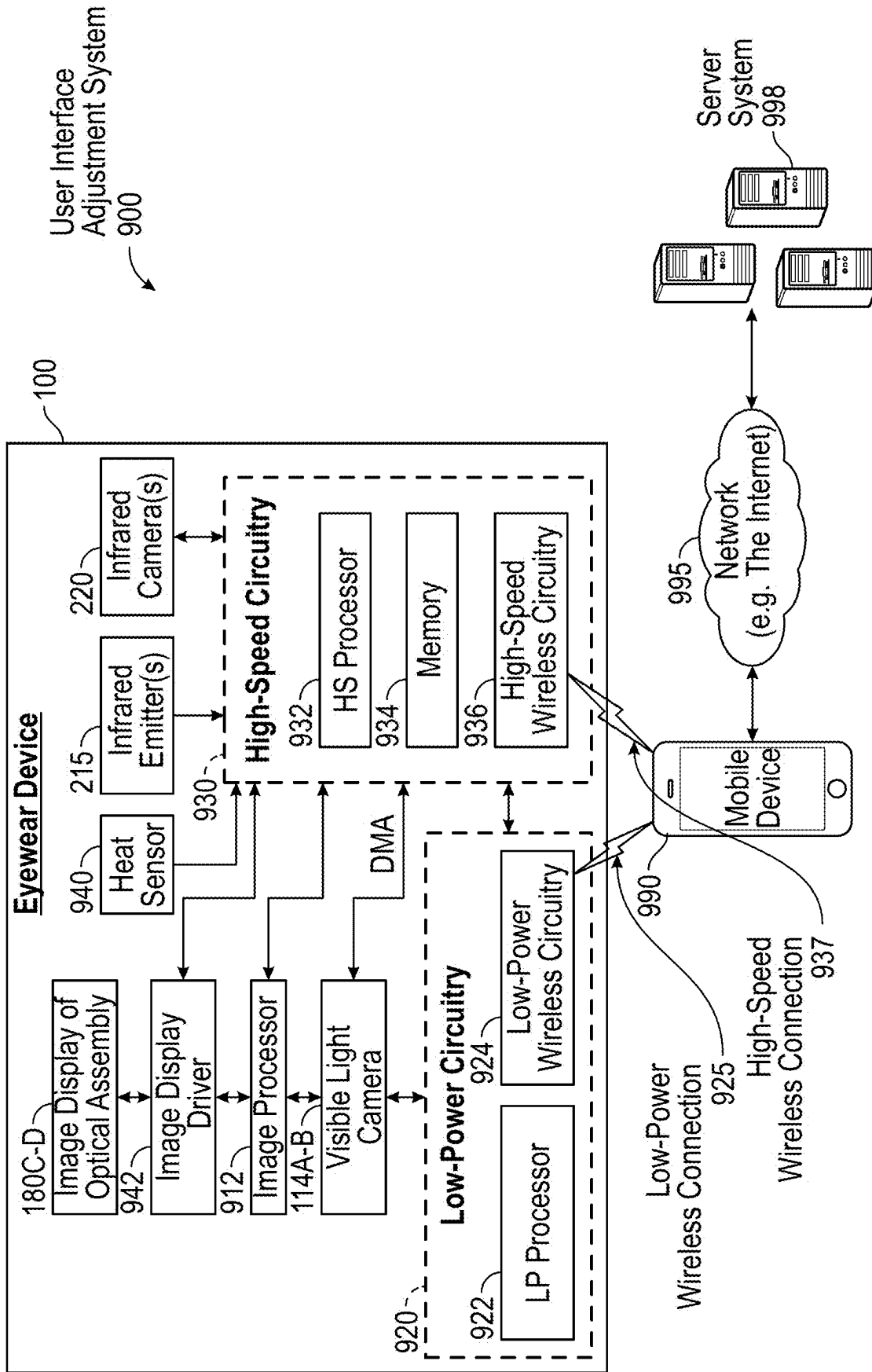


FIG. 9

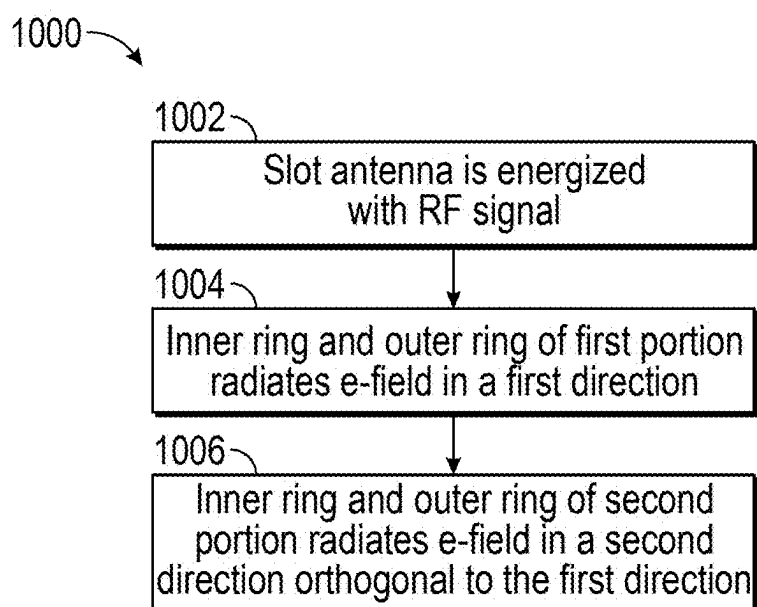


FIG. 10

EYEWEAR WITH SLOT-RING ANTENNA

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation of U.S. application Ser. No. 18/583,678 filed on Feb. 21, 2024, which is a Continuation of U.S. application Ser. No. 17/386,716 filed on Jul. 28, 2021, now U.S. Pat. No. 11,929,548, the contents of all of which are incorporated fully herein by reference.

TECHNICAL FIELD

[0002] The present subject matter relates to an eyewear device, e.g., smart glasses having see-through displays.

BACKGROUND

[0003] Portable eyewear devices, such as smart glasses, headwear, and headgear available today integrate cameras, see-through displays, and antennas.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The drawing figures depict one or more implementations, by way of example only, not by way of limitations. In the figures, like reference numerals refer to the same or similar elements.

[0005] FIG. 1A is a side view of an example hardware configuration of an eyewear device, which shows a right optical assembly with an image display, and field of view adjustments are applied to a user interface presented on the image display based on detected head or eye movement by a user;

[0006] FIG. 1B is a top cross-sectional view of a temple of the eyewear device of FIG. 1A depicting a visible light camera, a head movement tracker for tracking the head movement of the user of the eyewear device, and a circuit board;

[0007] FIG. 2A is a rear view of an example hardware configuration of an eyewear device, which includes an eye scanner on a frame, for use in a system for identifying a user of the eyewear device;

[0008] FIG. 2B is a rear view of an example hardware configuration of another eyewear device, which includes an eye scanner on a temple, for use in a system for identifying a user of the eyewear device;

[0009] FIGS. 2C and 2D are rear views of example hardware configurations of the eyewear device, including two different types of image displays.

[0010] FIG. 3 shows a rear perspective view of the eyewear device of FIG. 2A depicting an infrared emitter, an infrared camera, a frame front, a frame back, and a circuit board;

[0011] FIG. 4 is a cross-sectional view taken through the infrared emitter and the frame of the eyewear device of FIG. 3;

[0012] FIG. 5 illustrates detecting eye gaze direction;

[0013] FIG. 6 illustrates detecting eye position;

[0014] FIG. 7 depicts an example of visible light captured by the left visible light camera as a left raw image and visible light captured by the right visible light camera as a right raw image;

[0015] FIG. 8A illustrates a slot antenna created by a rectangular cutout, referred to as a slot;

[0016] FIG. 8B illustrates the E-field distribution of the slot antenna when the slot length, L , is less than (or equal to) half wavelength;

[0017] FIG. 8C illustrates a slot antenna shaped as a ring;

[0018] FIG. 8D illustrates overall size of the ring slot antenna of FIG. 8C is substantially smaller than the rectangular antenna of FIG. 8A;

[0019] FIG. 8E illustrates expanding the inner opening of the slot antenna;

[0020] FIG. 8F illustrates the ring slot design including changes in the shape of the design from a circle to a rectangle;

[0021] FIG. 8G illustrates a cutout in the center of the conductive material creating an inner conductor shaped as a ring;

[0022] FIG. 8H illustrates the inner conductor is sufficiently large enough such that additional E-field vectors extend from an inner edge of the inner conductor to an inner edge of the outer conductor;

[0023] FIG. 8I illustrates the antenna can be made to support more bands by adding a series LC circuit along the aperture and effectively change the total slot length at the resonant frequency;

[0024] FIG. 8J illustrates a voltage standing wave ratio (VSWR) for the slot antenna;

[0025] FIG. 8K illustrates where the size of the outer conductor is reduced to save space in the overall size of the antenna;

[0026] FIG. 8L illustrates an antenna having two slot-ring antennas that co-exist in very close proximity to each other and support the same frequency band, with minimal coupling to each other;

[0027] FIG. 8M illustrates the antenna of FIG. 8L with a narrowed portion between the two slot-ring antennas;

[0028] FIG. 8N illustrates the antenna of FIG. 8L integrated into eyewear;

[0029] FIG. 9 illustrates a block diagram of electronic components of the eyewear device; and

[0030] FIG. 10 is a flowchart of the operation of the double slot-ring antenna generating E-fields that are orthogonal to each other.

DETAILED DESCRIPTION

[0031] Examples described herein include eyewear with a slot ring antenna generating E-fields that are orthogonal to each other at a frequency. The slot ring antenna has a first portion including an inner ring and an outer ring configured to generate a first E-field in a first direction, and a second portion having a respective inner ring and the outer ring configured to generate a second E-field in a second direction that is orthogonal to the first direction. The outer ring encompasses the respective inner rings, where a first slot is defined between the respective inner ring and the outer ring. A second slot is encompassed by the respective inner ring, and it may be a cutout. The slot ring antenna encompasses the eyewear optical assemblies. In one example, the slot antenna first portion encompasses a first optical assembly, and the second portion encompasses the second optical assembly. The outer ring may comprise of a common ground plane, which ground plane may extend proximate a bridge of the eyewear. The two portions of the antenna co-exist in very close proximity to each other and support the same frequency band with minimal coupling to each other due to the orthogonal E-fields.

[0032] Additional objects, advantages and novel features of the examples will be set forth in part in the description which follows, and in part will become apparent to those skilled in the art upon examination of the following and the accompanying drawings or may be learned by production or operation of the examples. The objects and advantages of the present subject matter may be realized and attained by means of the methodologies, instrumentalities and combinations particularly pointed out in the appended claims.

[0033] In the following detailed description, numerous specific details are set forth by way of examples in order to provide a thorough understanding of the relevant teachings. However, it should be apparent to those skilled in the art that the present teachings may be practiced without such details. In other instances, well known methods, procedures, components, and circuitry have been described at a relatively high-level, without detail, in order to avoid unnecessarily obscuring aspects of the present teachings.

[0034] The term “coupled” as used herein refers to any logical, optical, physical or electrical connection, link or the like by which signals or light produced or supplied by one system element are imparted to another coupled element. Unless described otherwise, coupled elements or devices are not necessarily directly connected to one another and may be separated by intermediate components, elements or communication media that may modify, manipulate or carry the light or signals.

[0035] The orientations of the eyewear device, associated components and any complete devices incorporating an eye scanner and camera such as shown in any of the drawings, are given by way of example only, for illustration and discussion purposes. In operation for a particular variable optical processing application, the eyewear device may be oriented in any other direction suitable to the particular application of the eyewear device, for example up, down, sideways, or any other orientation. Also, to the extent used herein, any directional term, such as front, rear, inwards, outwards, towards, left, right, lateral, longitudinal, up, down, upper, lower, top, bottom and side, are used by way of example only, and are not limiting as to direction or orientation of any optic or component of an optic constructed as otherwise described herein.

[0036] Reference now is made in detail to the examples illustrated in the accompanying drawings and discussed below.

[0037] FIG. 1A is a side view of an example hardware configuration of an eyewear device 100, which includes a right optical assembly 180B with an image display 180D (FIG. 2A). Eyewear device 100 includes multiple visible light cameras 114A-B (FIG. 7) that form a stereo camera, of which the right visible light camera 114B is located on a right temple 110B.

[0038] The left and right visible light cameras 114A-B have an image sensor that is sensitive to the visible light range wavelength. Each of the visible light cameras 114A-B have a different frontward facing angle of coverage, for example, visible light camera 114B has the depicted angle of coverage 111B. The angle of coverage is an angle range which the image sensor of the visible light camera 114A-B picks up electromagnetic radiation and generates images. Examples of such visible light camera 114A-B include a high-resolution complementary metal-oxide-semiconductor (CMOS) image sensor and a video graphic array (VGA) camera, such as 640p (e.g., 640×480 pixels for a total of 0.3

megapixels), 720p, or 1080p. Image sensor data from the visible light cameras 114A-B are captured along with geo-location data, digitized by an image processor, and stored in a memory.

[0039] To provide stereoscopic vision, visible light cameras 114A-B may be coupled to an image processor (element 912 of FIG. 9) for digital processing along with a timestamp in which the image of the scene is captured. Image processor 912 includes circuitry to receive signals from the visible light camera 114A-B and process those signals from the visible light cameras 114A-B into a format suitable for storage in the memory (element 934 of FIG. 9). The timestamp can be added by the image processor 912 or other processor, which controls operation of the visible light cameras 114A-B. Visible light cameras 114A-B allow the stereo camera to simulate human binocular vision. Stereo cameras provide the ability to reproduce three-dimensional images (element 715 of FIG. 7) based on two captured images (elements 758A-B of FIG. 7) from the visible light cameras 114A-B, respectively, having the same timestamp. Such three-dimensional images 715 allow for an immersive life-like experience, e.g., for virtual reality or video gaming. For stereoscopic vision, the pair of images 758A-B are generated at a given moment in time—one image for each of the left and right visible light cameras 114A-B. When the pair of generated images 758A-B from the frontward facing angles of coverage 111A-B of the left and right visible light cameras 114A-B are stitched together (e.g., by the image processor 912), depth perception is provided by the optical assembly 180A-B.

[0040] In an example, a user interface field of view adjustment system includes the eyewear device 100. The eyewear device 100 includes a frame 105, a right temple 110B extending from a right lateral side 170B of the frame 105, and a see-through image display 180D (FIGS. 2A-B) comprising optical assembly 180B to present a graphical user interface to a user. The eyewear device 100 includes the left visible light camera 114A connected to the frame 105 or the left temple 110A to capture a first image of the scene. Eyewear device 100 further includes the right visible light camera 114B connected to the frame 105 or the right temple 110B to capture (e.g., simultaneously with the left visible light camera 114A) a second image of the scene which partially overlaps the first image. Although not shown in FIGS. 1A-B, the user interface field of view adjustment system further includes the processor 932 coupled to the eyewear device 100 and connected to the visible light cameras 114A-B, the memory 934 accessible to the processor 932, and programming in the memory 934, for example in the eyewear device 100 itself or another part of the user interface field of view adjustment system.

[0041] Although not shown in FIG. 1A, the eyewear device 100 also includes a head movement tracker (element 109 of FIG. 1B) or an eye movement tracker (element 213 of FIGS. 2A-B). Eyewear device 100 further includes the see-through image displays 180C-D of optical assembly 180A-B for presenting a sequence of displayed images, and an image display driver (element 942 of FIG. 9) coupled to the see-through image displays 180C-D of optical assembly 180A-B to control the image displays 180C-D of optical assembly 180A-B to present the sequence of displayed images 715, which are described in further detail below. Eyewear device 100 further includes the memory 934 and the processor 932 having access to the image display driver

942 and the memory **934**. Eyewear device **100** further includes programming (element **934** of FIG. 9) in the memory. Execution of the programming by the processor **932** configures the eyewear device **100** to perform functions, including functions to present, via the see-through image displays **180C-D**, an initial displayed image of the sequence of displayed images, the initial displayed image having an initial field of view corresponding to an initial head direction or an initial eye gaze direction (element **230** of FIG. 5).

[0042] Execution of the programming by the processor **932** further configures the eyewear device **100** to detect movement of a user of the eyewear device by: (i) tracking, via the head movement tracker (element **109** of FIG. 1B), a head movement of a head of the user, or (ii) tracking, via an eye movement tracker (element **113**, **213** of FIGS. 2A-B, FIG. 5), an eye movement of an eye of the user of the eyewear device **100**. Execution of the programming by the processor **932** further configures the eyewear device **100** to determine a field of view adjustment to the initial field of view of the initial displayed image based on the detected movement of the user. The field of view adjustment includes a successive field of view corresponding to a successive head direction or a successive eye direction. Execution of the programming by the processor **932** further configures the eyewear device **100** to generate a successive displayed image of the sequence of displayed images based on the field of view adjustment. Execution of the programming by the processor **932** further configures the eyewear device **100** to present, via the see-through image displays **180C-D** of the optical assembly **180A-B**, the successive displayed images.

[0043] FIG. 1B is a top cross-sectional view of the temple of the eyewear device **100** of FIG. 1A depicting the right visible light camera **114B**, a head movement tracker **109**, and a circuit board. Construction and placement of the left visible light camera **114A** is substantially similar to the right visible light camera **114B**, except the connections and coupling are on the left lateral side **170A**. As shown, the eyewear device **100** includes the right visible light camera **114B**, a circuit board, which may be a flexible printed circuit board (PCB) **140**, and a battery configured to power eyewear **100**. The right hinge **226B** connects the right temple **110B** to a right temple **125B** of the eyewear device **100**. In some examples, components of the right visible light camera **114B**, the flexible PCB **140**, or other electrical connectors or contacts may be located on the right temple **125B** or the right hinge **226B**.

[0044] As shown, eyewear device **100** has a head movement tracker **109**, which includes, for example, an inertial measurement unit (IMU). An inertial measurement unit is an electronic device that measures and reports a body's specific force, angular rate, and sometimes the magnetic field surrounding the body, using a combination of accelerometers and gyroscopes, sometimes also magnetometers. The inertial measurement unit works by detecting linear acceleration using one or more accelerometers and rotational rate using one or more gyroscopes. Typical configurations of inertial measurement units contain one accelerometer, gyro, and magnetometer per axis for each of the three axes: horizontal axis for left-right movement (X), vertical axis (Y) for top-bottom movement, and depth or distance axis for up-down movement (Z). The accelerometer detects the gravity vector. The magnetometer defines the rotation in the magnetic field (e.g., facing south, north, etc.) like a compass which generates a heading reference. The three accelerom-

eters to detect acceleration along the horizontal, vertical, and depth axis defined above, which can be defined relative to the ground, the eyewear device **100**, or the user wearing the eyewear device **100**.

[0045] Eyewear device **100** detects movement of the user of the eyewear device **100** by tracking, via the head movement tracker **109**, the head movement of the head of the user. The head movement includes a variation of head direction on a horizontal axis, a vertical axis, or a combination thereof from the initial head direction during presentation of the initial displayed image on the image display. In one example, tracking, via the head movement tracker **109**, the head movement of the head of the user includes measuring, via the inertial measurement unit **109**, the initial head direction on the horizontal axis (e.g., X axis), the vertical axis (e.g., Y axis), or the combination thereof (e.g., transverse or diagonal movement). Tracking, via the head movement tracker **109**, the head movement of the head of the user further includes measuring, via the inertial measurement unit **109**, a successive head direction on the horizontal axis, the vertical axis, or the combination thereof during presentation of the initial displayed image.

[0046] Tracking, via the head movement tracker **109**, the head movement of the head of the user further includes determining the variation of head direction based on both the initial head direction and the successive head direction. Detecting movement of the user of the eyewear device **100** further includes in response to tracking, via the head movement tracker **109**, the head movement of the head of the user, determining that the variation of head direction exceeds a deviation angle threshold on the horizontal axis, the vertical axis, or the combination thereof. The deviation angle threshold is between about 3° to 10°. As used herein, the term “about” when referring to an angle means $\pm 10\%$ from the stated amount.

[0047] Variation along the horizontal axis slides three-dimensional objects, such as characters, Bitmojis, application icons, etc. in and out of the field of view by, for example, hiding, unhiding, or otherwise adjusting visibility of the three-dimensional object. Variation along the vertical axis, for example, when the user looks upwards, in one example, displays weather information, time of day, date, calendar appointments, etc. In another example, when the user looks downwards on the vertical axis, the eyewear device **100** may power down.

[0048] The right temple **110B** includes temple body **211** and a temple cap, with the temple cap omitted in the cross-section of FIG. 1B. Disposed inside the right temple **110B** are various interconnected circuit boards, such as PCBs or flexible PCBs, that include controller circuits for right visible light camera **114B**, microphone(s) **130**, speaker(s) **132**, low-power wireless circuitry and antennas (e.g., for wireless short-range network communication via Bluetooth™), and high-speed wireless circuitry and antennas (e.g., for wireless local area network communication via WiFi and positioning via GPS).

[0049] The right visible light camera **114B** is coupled to or disposed on the flexible PCB **40** and covered by a visible light camera cover lens, which is aimed through opening(s) formed in the right temple **110B**. In some examples, the frame **105** connected to the right temple **110B** includes the opening(s) for the visible light camera cover lens. The frame **105** includes a front-facing side configured to face outwards away from the eye of the user. The opening for the visible

light camera cover lens is formed on and through the front-facing side. In the example, the right visible light camera 114B has an outward facing angle of coverage 111B with a line of sight or perspective of the right eye of the user of the eyewear device 100. The visible light camera cover lens can also be adhered to an outward facing surface of the right temple 110B in which an opening is formed with an outwards facing angle of coverage, but in a different outwards direction. The coupling can also be indirect via intervening components.

[0050] Left (first) visible light camera 114A is connected to the left see-through image display 180C of left optical assembly 180A to generate a first background scene of a first successive displayed image. The right (second) visible light camera 114B is connected to the right see-through image display 180D of right optical assembly 180B to generate a second background scene of a second successive displayed image. The first background scene and the second background scene partially overlap to present a three-dimensional observable area of the successive displayed image.

[0051] Flexible PCB 140 is disposed inside the right temple 110B and is coupled to one or more other components housed in the right temple 110B. Although shown as being formed on the circuit boards of the right temple 110B, the right visible light camera 114B can be formed on the circuit boards of the left temple 110A, the temples 125A-B, or frame 105.

[0052] FIG. 2A is a rear view of an example hardware configuration of an eyewear device 100, which includes an eye scanner 113 on a frame 105, for use in a system for determining an eye position and gaze direction of a wearer/user of the eyewear device 100. As shown in FIG. 2A, the eyewear device 100 is in a form configured for wearing by a user, which are eyeglasses in the example of FIG. 2A. The eyewear device 100 can take other forms and may incorporate other types of frameworks, for example, a headgear, a headset, or a helmet.

[0053] In the eyeglasses example, eyewear device 100 includes the frame 105 which includes the left rim 107A connected to the right rim 107B via the bridge 106 adapted for a nose of the user. The left and right rims 107A-B include respective apertures 175A-B which hold the respective optical element 180A-B, such as a lens and the see-through displays 180C-D. As used herein, the term lens is meant to cover transparent or translucent pieces of glass or plastic having curved and flat surfaces that cause light to converge/diverge or that cause little or no convergence/divergence.

[0054] Although shown as having two optical elements 180A-B, the eyewear device 100 can include other arrangements, such as a single optical element depending on the application or intended user of the eyewear device 100. As further shown, eyewear device 100 includes the left temple 110A adjacent the left lateral side 170A of the frame 105 and the right temple 110B adjacent the right lateral side 170B of the frame 105. The temples 110A-B may be integrated into the frame 105 on the respective sides 170A-B (as illustrated) or implemented as separate components attached to the frame 105 on the respective sides 170A-B. Alternatively, the temples 110A-B may be integrated into temples (not shown) attached to the frame 105.

[0055] In the example of FIG. 2A, the eye scanner 113 includes an infrared emitter 115 and an infrared camera 120. Visible light cameras typically include a blue light filter to block infrared light detection, in an example, the infrared

camera 120 is a visible light camera, such as a low-resolution video graphic array (VGA) camera (e.g., 640×480 pixels for a total of 0.3 megapixels), with the blue filter removed. The infrared emitter 115 and the infrared camera 120 are co-located on the frame 105, for example, both are shown as connected to the upper portion of the left rim 107A. The frame 105 or one or more of the left and right temples 110A-B include a circuit board (not shown) that includes the infrared emitter 115 and the infrared camera 120. The infrared emitter 115 and the infrared camera 120 can be connected to the circuit board by soldering, for example.

[0056] Other arrangements of the infrared emitter 115 and infrared camera 120 can be implemented, including arrangements in which the infrared emitter 115 and infrared camera 120 are both on the right rim 107B, or in different locations on the frame 105, for example, the infrared emitter 115 is on the left rim 107A and the infrared camera 120 is on the right rim 107B. In another example, the infrared emitter 115 is on the frame 105 and the infrared camera 120 is on one of the temples 110A-B, or vice versa. The infrared emitter 115 can be connected essentially anywhere on the frame 105, left temple 110A, or right temple 110B to emit a pattern of infrared light. Similarly, the infrared camera 120 can be connected essentially anywhere on the frame 105, left temple 110A, or right temple 110B to capture at least one reflection variation in the emitted pattern of infrared light.

[0057] The infrared emitter 115 and infrared camera 120 are arranged to face inwards towards an eye of the user with a partial or full field of view of the eye in order to identify the respective eye position and gaze direction. For example, the infrared emitter 115 and infrared camera 120 are positioned directly in front of the eye, in the upper part of the frame 105 or in the temples 110A-B at either ends of the frame 105.

[0058] FIG. 2B is a rear view of an example hardware configuration of another eyewear device 200. In this example configuration, the eyewear device 200 is depicted as including an eye scanner 213 on a right temple 210B. As shown, an infrared emitter 215 and an infrared camera 220 are co-located on the right temple 210B. It should be understood that the eye scanner 213 or one or more components of the eye scanner 213 can be located on the left temple 210A and other locations of the eyewear device 200, for example, the frame 105. The infrared emitter 215 and infrared camera 220 are like that of FIG. 2A, but the eye scanner 213 can be varied to be sensitive to different light wavelengths as described previously in FIG. 2A.

[0059] Similar to FIG. 2A, the eyewear device 200 includes a frame 105 which includes a left rim 107A which is connected to a right rim 107B via a bridge 106; and the left and right rims 107A-B include respective apertures which hold the respective optical elements 180A-B comprising the see-through display 180C-D.

[0060] FIGS. 2C-D are rear views of example hardware configurations of the eyewear device 100, including two different types of see-through image displays 180C-D. In one example, these see-through image displays 180C-D of optical assembly 180A-B include an integrated image display. As shown in FIG. 2C, the optical assemblies 180A-B includes a suitable display matrix 180C-D of any suitable type, such as a liquid crystal display (LCD), an organic light-emitting diode (OLED) display, a waveguide display, or any other such display. The optical assembly 180A-B also

includes an optical layer or layers **176**, which can include lenses, optical coatings, prisms, mirrors, waveguides, optical strips, and other optical components in any combination. The optical layers **176A-N** can include a prism having a suitable size and configuration and including a first surface for receiving light from display matrix and a second surface for emitting light to the eye of the user. The prism of the optical layers **176A-N** extends over all or at least a portion of the respective apertures **175A-B** formed in the left and right rims **107A-B** to permit the user to see the second surface of the prism when the eye of the user is viewing through the corresponding left and right rims **107A-B**. The first surface of the prism of the optical layers **176A-N** faces upwardly from the frame **105** and the display matrix overlies the prism so that photons and light emitted by the display matrix impinge the first surface. The prism is sized and shaped so that the light is refracted within the prism and is directed towards the eye of the user by the second surface of the prism of the optical layers **176A-N**. In this regard, the second surface of the prism of the optical layers **176A-N** can be convex to direct the light towards the center of the eye. The prism can optionally be sized and shaped to magnify the image projected by the see-through image displays **180C-D**, and the light travels through the prism so that the image viewed from the second surface is larger in one or more dimensions than the image emitted from the see-through image displays **180C-D**.

[0061] In another example, the see-through image displays **180C-D** of optical assembly **180A-B** include a projection image display as shown in FIG. 2D. The optical assembly **180A-B** includes a laser projector **150**, which is a three-color laser projector using a scanning mirror or galvanometer. During operation, an optical source such as a laser projector **150** is disposed in or on one of the temples **125A-B** of the eyewear device **100**. Optical assembly **180A-B** includes one or more optical strips **155A-N** spaced apart across the width of the lens of the optical assembly **180A-B** or across a depth of the lens between the front surface and the rear surface of the lens.

[0062] As the photons projected by the laser projector **150** travel across the lens of the optical assembly **180A-B**, the photons encounter the optical strips **155A-N**. When a particular photon encounters a particular optical strip, the photon is either redirected towards the user's eye, or it passes to the next optical strip. A combination of modulation of laser projector **150**, and modulation of optical strips, may control specific photons or beams of light. In an example, a processor controls optical strips **155A-N** by initiating mechanical, acoustic, or electromagnetic signals. Although shown as having two optical assemblies **180A-B**, the eyewear device **100** can include other arrangements, such as a single or three optical assemblies, or the optical assembly **180A-B** may have arranged different arrangement depending on the application or intended user of the eyewear device **100**.

[0063] As further shown in FIGS. 2C-D, eyewear device **100** includes a left temple **110A** adjacent the left lateral side **170A** of the frame **105** and a right temple **110B** adjacent the right lateral side **170B** of the frame **105**. The temples **110A-B** may be integrated into the frame **105** on the respective lateral sides **170A-B** (as illustrated) or implemented as separate components attached to the frame **105** on

the respective sides **170A-B**. Alternatively, the temples **110A-B** may be integrated into temples **125A-B** attached to the frame **105**.

[0064] In one example, the see-through image displays include the first see-through image display **180C** and the second see-through image display **180D**. Eyewear device **100** includes first and second apertures **175A-B** which hold the respective first and second optical assembly **180A-B**. The first optical assembly **180A** includes the first see-through image display **180C** (e.g., a display matrix of FIG. 2C or optical strips **155A-N'** and a projector **150A**). The second optical assembly **180B** includes the second see-through image display **180D** e.g., a display matrix of FIG. 2C or optical strips **155A-N''** and a projector **150B**). The successive field of view of the successive displayed image includes an angle of view between about 15° to 30°, and more specifically 24°, measured horizontally, vertically, or diagonally. The successive displayed image having the successive field of view represents a combined three-dimensional observable area visible through stitching together of two displayed images presented on the first and second image displays.

[0065] As used herein, “an angle of view” describes the angular extent of the field of view associated with the displayed images presented on each of the left and right image displays **180C-D** of optical assembly **180A-B**. The “angle of coverage” describes the angle range that a lens of visible light cameras **114A-B** or infrared camera **220** can image. Typically, the image circle produced by a lens is large enough to cover the film or sensor completely, possibly including some vignetting (i.e., a reduction of an image's brightness or saturation toward the periphery compared to the image center). If the angle of coverage of the lens does not fill the sensor, the image circle will be visible, typically with strong vignetting toward the edge, and the effective angle of view will be limited to the angle of coverage. The “field of view” is intended to describe the field of observable area which the user of the eyewear device **100** can see through his or her eyes via the displayed images presented on the left and right image displays **180C-D** of the optical assembly **180A-B**. Image display **180C** of optical assembly **180A-B** can have a field of view with an angle of coverage between 15° to 30°, for example 24°, and have a resolution of 480×480 pixels.

[0066] FIG. 3 shows a rear perspective view of the eyewear device of FIG. 2A. The eyewear device **100** includes an infrared emitter **215**, infrared camera **220**, a frame front **330**, a frame back **335**, and a circuit board **340**. It can be seen in FIG. 3 that the upper portion of the left rim of the frame of the eyewear device **100** includes the frame front **330** and the frame back **335**. An opening for the infrared emitter **215** is formed on the frame back **335**.

[0067] As shown in the encircled cross-section **4** in the upper middle portion of the left rim of the frame, a circuit board, which is a flexible PCB **340**, is sandwiched between the frame front **330** and the frame back **335**. Also shown in further detail is the attachment of the left temple **110A** to the left temple **325A** via the left hinge **326A**. In some examples, components of the eye movement tracker **213**, including the infrared emitter **215**, the flexible PCB **340**, or other electrical connectors or contacts may be located on the left temple **325A** or the left hinge **326A**.

[0068] FIG. 4 is a cross-sectional view through the infrared emitter **215** and the frame corresponding to the encircled

cross-section 4 of the eyewear device of FIG. 3. Multiple layers of the eyewear device 100 are illustrated in the cross-section of FIG. 4, as shown the frame includes the frame front 330 and the frame back 335. The flexible PCB 340 is disposed on the frame front 330 and connected to the frame back 335. The infrared emitter 215 is disposed on the flexible PCB 340 and covered by an infrared emitter cover lens 445. For example, the infrared emitter 215 is reflowed to the back of the flexible PCB 340. Reflowing attaches the infrared emitter 215 to contact pad(s) formed on the back of the flexible PCB 340 by subjecting the flexible PCB 340 to controlled heat which melts a solder paste to connect the two components. In one example, reflowing is used to surface mount the infrared emitter 215 on the flexible PCB 340 and electrically connect the two components. However, it should be understood that through-holes can be used to connect leads from the infrared emitter 215 to the flexible PCB 340 via interconnects, for example.

[0069] The frame back 335 includes an infrared emitter opening 450 for the infrared emitter cover lens 445. The infrared emitter opening 450 is formed on a rear-facing side of the frame back 335 that is configured to face inwards towards the eye of the user. In the example, the flexible PCB 340 can be connected to the frame front 330 via the flexible PCB adhesive 460. The infrared emitter cover lens 445 can be connected to the frame back 335 via infrared emitter cover lens adhesive 455. The coupling can also be indirect via intervening components.

[0070] In an example, the processor 932 utilizes eye tracker 213 to determine an eye gaze direction 230 of a wearer's eye 234 as shown in FIG. 5, and an eye position 236 of the wearer's eye 234 within an eyebox as shown in FIG. 6. The eye tracker 213 is a scanner which uses infrared light illumination (e.g., near-infrared, short-wavelength infrared, mid-wavelength infrared, long-wavelength infrared, or far infrared) to captured image of reflection variations of infrared light from the eye 234 to determine the gaze direction 230 of a pupil 232 of the eye 234, and also the eye position 236 with respect to the see-through display 180D.

[0071] FIG. 7 depicts an example of capturing visible light with cameras. Visible light is captured by the left visible light camera 114A with a left visible light camera field of view 111A as a left raw image 758A. Visible light is captured by the right visible light camera 114B with a right visible light camera field of view 111B as a right raw image 758B having an overlap 713 with the left field of view 111A. Based on processing of the left raw image 758A and the right raw image 758B, a three-dimensional depth map 715 of a three-dimensional scene, referred to hereafter as an image, is generated by processor 932.

Slot-Ring Antenna

[0072] Augmented reality (AR) devices enable mixing computer-generated imagery and audio with the real world, possibly in variable proportions, to achieve an unmatched level of immersion. AR technology is beneficial for many applications from changing maps behind weather reporters to displays embedded in the windshields of a car to show telemetry and navigational data. This technology is particularly compelling when it is wearable, such as smart glasses and eyewear. When worn, this AR technology can interactively present images to the wearer's eye and sounds to the wearer's ears on demand in an instant.

[0073] Incorporating AR in a wearable form factor is a difficult task, as this technology requires an immense amount of computational power to "map" the environment. If all the processing is done on the wearable device itself, these powerful processors require energy-dense batteries to electrify them and excellent thermal mitigation to ensure it is efficient and comfortable to wear. The optical engine needs to be very efficient in turning electricity into light and projecting it to the wearers' eye. There is also a fashion angle, as wearable devices have to be lightweight, look good when worn, and have excellent weight distribution to gain mainstream appeal.

[0074] A low latency wireless communication system integrated into the wearable AR device helps alleviate some of the difficulty by offloading a portion of the computational work to a remote computer, such as a puck, a nearby smartphone, or a cloud service. 5G New Radio (NR) and WiFi 6 offer extremely low latency communication schemes, however, they require multiple multi-band highly efficient orthogonal antenna designs to function correctly. Even in the scenario where computing can be done on-board the wearable device, a low latency, high speed connection to wide area network (WAN) can meaningfully expand the capabilities of the wearable AR device, from downloading the pre-mapped environments to live streaming content. It is desirable to enable this wireless connectivity in an antenna architecture that works efficiently when integrated into the AR wearable device.

[0075] It is critical that the wireless communication systems embedded into the AR wearables are highly efficient, immune to detune and de-sense when worn on head, all the while complying with stringent safety regulations such as specific absorption rate (SAR). As such, there is a need for metal free physical volume on smart glasses to place these antennas and accomplish performance goals. However, as mentioned earlier, space and weight come at a very large premium in these devices as they are also fashion accessories. Hence, there is a big incentive to reuse an existing volume for antenna design.

[0076] One physically large, nonmetallic structure in the AR-enabled eyewear with RF friendly material properties is an optical waveguide communicating light, such as an image from an image display or non-visible light for eye tracking. Leveraging the frame that holds the waveguide in place as an antenna is one way of enabling high speed wireless connectivity on these devices. There might be multiple ways of leveraging the frame structure and the waveguide that it encapsulates as an antenna. Slot-ring concept could be the most appropriate for the wearable application for its flexibility in achieving multi band radiation, and on body efficiency.

[0077] A good starting point for understanding the fundamentals of slot-ring antennas is to understand how the slot antenna works. As seen in FIG. 8A, a slot antenna 800 is formed by creating a rectangular cutout 802, referred to as a slot, in a conductive material 804, such as a metal, and placing an antenna feed 806 across the slot opening as shown. The width of the slot 802, h , is typically much smaller than the length of the slot, L .

[0078] FIG. 8B shows the E-field distribution of the slot antenna 800 when the slot length, L , is less than (or equal to) half wavelength. The E-fields peak at the center of the slot 802 and get sinusoidally diminished as they move towards the both edges, as the slot edges are essentially shorted

(since $h \ll L$) and results in creation of no E-field. The field vectors for the E-field are all aligned and point in the same direction, implying constructive time-varying E-fields, as shown FIG. 8B by the arrows. As a result of this behavior, radiation is generated from the slot opening **802**. The location of the antenna feed **806** determines an input impedance this antenna presents to an RF front end. The closer the antenna **800** is to the edges, the lower the impedance, and the closer it is to the center, the higher the impedance.

[0079] Referring to FIG. 8C, a slot antenna **810** is shown as a ring. The width of the slot antenna **810** is still h , and the length of the slot, now the periphery of the ring, is still L . The two shorting ends in FIG. 8C now merge and form a single short at **808**, connecting the conductive material **804** inside the slot **802** to the conductive material **804** outside the slot **802**.

[0080] FIG. 8C shows the E-field distribution of the slot-ring antenna **810** when the slot length, which is the ring periphery, L , is less than (or equal to) half wavelength. The E-fields have a similar distribution to the original slot antenna **800**, but the E-fields follow the curvature of the ring. E-fields peak right across the “Ring Short” location **808**, as that is the equi-distance point from both ends of the shorts in the original slot **802**. As seen on FIG. 8C, the E-field vectors do not always constructively add up as neatly and there are some cancellations of energy. This means the ring-slot antenna **810** in this form is a poorer antenna than a regular slot antenna **800**, but the overall size of the slot antenna is substantially smaller, as shown in FIG. 8D.

Eyewear Including a Slot-Ring Antenna

[0081] The ring slot antenna **800** and **810** radiate effectively by leveraging the E-field vectors in the slot that add constructively. The RF energy is mostly confined to the slot **802** and there is minimal current at the center of the inner conductive material **804**. As such, there is very little to lose by shaving it off a bit to create opening **822**, as shown in FIG. 8E, to create slot antenna **820**. As long as the $r2 \ll r1$ condition holds, the antenna behavior remains unchanged.

[0082] Referring to FIG. 8F, the ring slot design is improved by changing the shape of the design from a circle to a rectangle. As seen, the E-field vectors are no longer semi-cancelling, but instead, add constructively where it matters the most. There are some opposing E-field vectors on the short edges of the rectangular cutout **802** that will cancel, but since the majority of the E-field is located on the long edge that is across a shorting pin, the overall radiation efficiency of the antenna **830** is meaningfully higher than antenna **820** shown in FIG. 8E. A small opening **832** is created in the center of the ring antenna as shown.

[0083] Referring to FIG. 8G, antenna **830** is improved to generate antenna **840** by increasing ring dimension $r2$ such that the $r2 \ll r1$ condition no longer holds. A cutout **842** in the center of the conductive material **804** creates an inner conductor **844**, shaped as a ring. This inner conductor **844** is sufficiently large enough such that additional E-field vectors **846** extend from an inner edge **848** of the inner conductor **844** to an inner edge **850** of the outer conductor, shown as the conductive material **804** in FIG. 8H. The outer conductive material **804** is also referred to as the ground plane. These E-field vectors **846** are in addition to the original E-Field vectors **852** that extend from an outer edge **854** of the inner conductor **844** to the inner edge **850** of the outer conductive material **804**. The tapering of the amplitude of

the E-field vectors **846** and **852** of the antenna **840** are exactly the same as the tapering of original E-field vectors shown in FIG. 8F. The amplitude of the E-field vectors **846** and **852** peak at the location opposite of a shorting pin **856** and gradually decrease as it moves away from that location.

[0084] The antenna **840** shown in FIG. 8H allows for the radiating E-fields to be essentially more distributed compared to the baseline antenna of FIG. 8F, which is equivalent to widening the slot **802** (i.e., increasing the effective h). Increased slot width, as long as $heff \ll L$ condition still holds, results in increased operating bandwidth for the slot antenna **840** as it reduces the extra capacitance the antenna was seeing from the slots.

[0085] The antenna **840** is flexible enough to operate in multiple frequencies. If $L = \lambda/2$ at frequency $f1$, it is straightforward to provide a feed location such that the antenna is resonant at frequency $f1$. Given the nature of the slot antenna, it will also be resonant at frequencies $3f1$, $5f1$ etc. As shown in FIG. 8I, the antenna **840** can be made to support more bands by adding a series LC circuit at **874** along the aperture and effectively change the total slot length at the resonant frequency of this series LC circuit, forming an aperture tuned antenna **860**. The aperture tuned slot-ring antenna **860** sees the series LC circuit as open at all frequency bands other than $f2$. When the operating frequency is lower than $f2$, the capacitor dominates and when the frequency is higher, the inductor starts dominating the impedance. In a frequency band around the resonant frequency, however, this circuitry is short, essentially a replica of the ring short. If $L2$ is chosen such that $L2 = \lambda/2$, where $\lambda/2$ is the wavelength at $f2$, then there is effectively two slot antennas shown in FIG. 8I. The first original one that has the total length L , and a second one that only appears at $f2$ with a total length of $L2$. A voltage standing wave ratio (VSWR) for this antenna **860** is shown in FIG. 8J. As seen, there is a natural resonance at $f1$ and $f2$ and the odd harmonics of $f1$. It is possible to merge the bands by changing the lengths accordingly to increase the bandwidth or cover more bands as shown in FIG. 8J. More of these tuned series LC circuits can be added to the aperture to increase the number of bands the antenna can support. The locations of these series LC circuits are chosen such that they are half-wavelength away from the ring short in the direction of the antenna feed at the resonant frequency.

Double Slot Antenna

[0086] Referring to FIG. 8K, there is shown antenna **870** where the size of the outer conductive material **804** is reduced to save space in the overall size of the antenna. In an ideal slot antenna, the outer conductive material **804** would be infinite, however, that is not practical. As such, typical slot antennas rely on electrically large ground planes ($>10\lambda$). Total E-field radiated by a slot antenna is the combination of the E-field from the slot **802** and one that is diffracted from the edges of the finite outer conductive material **804**. When the outer conductive material **804** is large, the diffracted field is negligible. As the outer conductive material **804** gets smaller, forming the outer ring, there is more energy that gets diffracted at the edge of the ground plane and this results in reduced antenna gain. However, for many consumer electronics applications, this is a price that should be paid to accommodate the antenna size, but every effort should be made to increase the width of the outer conductor as much as possible.

[0087] The overall size of antenna **870** can be further reduced by dielectric loading the cutout **842**, also referred to as an inner slot, and the outer slot **802**, with a fill material. The higher the permittivity the fill material is, the higher effective permittivity the slot antenna will see. If the fill material has conductivity, that translates into ohmic losses for the antenna, so it is critical to use as pure dielectric material as possible for maximum antenna efficiency. The resonant frequency of the antenna **870** is inversely proportional to the square root of the effective permittivity. As such, great antenna performance benefits can be extracted from the same physical volume by dielectric loading. The cost of reducing the size of the antenna **870** with this method is the reduction in fractional bandwidth. Dielectric loading concentrates the radiating E-fields by pulling them into it, which results in a narrower impedance bandwidth. However, combining the multi-band methods described earlier, the loss of bandwidth is compensated to a certain degree, while maintaining the miniaturization.

[0088] It is possible to fill the inner slot **842** with material used to make waveguide glass. These materials typically have very low loss tangent (<0.001) and have high relative permittivity (>10). The outer slot **802** can be filled with any low loss insulating material, such as plastic or even air depending on the construction of this shape. In addition, antenna **870** has its feed **872** and ring-short position **874** positioned 90 degrees offset from one another, creating radiating E-fields in the fundamental mode (at f1) that are orthogonal to the ones shown in FIG. 8H.

[0089] Referring to FIG. 8L, there is shown an antenna **880** having two slot-ring antennas **880A** and **880B** that co-exist in very close proximity to each other and support the same frequency band (in this example f1), with minimal coupling to each other because of the orthogonal E-fields. Antenna **880A** and **880B** have a feed **882A** and feed **882B**, respectively, and a ring-short position **884A** and **884B**, respectively, creating respective radiating E-fields in the fundamental mode (at f1) that are orthogonal to the ones shown in FIG. 8H. The coupling is frequency dependent as the radiating E-fields are not always constrained to the orthogonal edges between the two antennas (there might be increased coupling at f2 or 3 f1).

[0090] The fundamental antenna design of antenna **880** is an antenna array including antennas **880A** and **880B** each formed from slot-rings that support the two orthogonal E-field polarizations, thus ensuring meaningful polarization diversity for a wireless communication system that can be embedded inside a wearable augmented reality device, such as eyewear **100**.

[0091] Referring to FIG. 8M, the antenna **890** is antenna **880** modified such that it can be integrated into eyewear **100**. The portion of conductive material **804** extending between the antennas **880A** and **880B** is narrowed to correspond to the bridge of the frame **105** of the eyewear **100**, and the opposing far ends of the conductive material **804** are extended such that they can be integrated into the temple **110** of eyewear **100**. There may be extra precautions taken to improve the isolation between the array elements.

[0092] Referring to FIG. 8N, the antenna **890** is shown integrated into eyewear **100**. The slot-ring antenna **890** can be implemented in eyewear **100** in a number of ways. One method is to start with a block of aluminum, machine the aluminum into the shape of the frame **105** that is fashionable, light, and can carry optical elements **180A** and **180B** to

display images on the respective displays **180C** and **180D**. On this aluminum frame **105**, machine out the slots **802** with an antenna friendly gap and leave a recessed shorting bar in place in the correct position. Next, mold a polycarbonate insulator in the gap that is formed by machining out the slot **802** to form antennas **880A** and **880B**. This ensures the final frame **105** is smooth and fashionable. The optical elements **180A** and **180B** can be separated for each eye and the same treatment can be applied to both. A transceiver (not shown) for the antenna **890** can be located in the temple **110** and the energy can be transferred from the transceiver to the antenna **890** via coaxial cables, or a liquid crystal polymer (LCP) based transmission lines. As seen, this design makes both the inner conductor **844** and the outer conductive material **804** visible. With this design, the outer slot **802** becomes part of the industrial design and the polycarbonate material choice can be modified to fit fashion needs. It could also be painted.

[0093] Another method of implementing this antenna **890** is to use a metal frame **105** and leverage metal frame as the outer conductive material **804**. Then, for the inner conductor **844** uses a flexible printed circuit (FPC) based antenna that is adhered to the outer lens on its inner surface and hide the FPC antenna behind a bezel. This method enables both the inner slot **842** and the outer slot **802** to be filled with lens material essentially. In the same vein, the inner conductor **844** could be part of an assembly structure along with the waveguides and lenses and could be installed alongside the waveguides.

[0094] The antenna **890** can also be implemented if the frame **105** is plastic. In this case, both the inner and the outer conductors are molded inside the plastic, or they are inserted into a housing that is carved out of the plastic. In this implementation, the outer slot **802** is filled with the plastic used as the frame **105**, and the inner slot **842** is filled with the lens material.

[0095] FIG. 9 depicts a high-level functional block diagram including example electronic components disposed in eyewear **100/200**. The illustrated electronic components include the processor **932**, the memory **934**, and the see-through image display **180C** and **180D** including the embedded antennas **808**.

[0096] Memory **934** includes instructions for execution by processor **932** to implement functionality of eyewear **100/200**, including instructions for processor **932** to control in the image **715**. Processor **932** receives power from battery (not shown) and executes the instructions stored in memory **934**, or integrated with the processor **932** on-chip, to perform functionality of eyewear **100/200**, and communicating with external devices via wireless connections.

[0097] A user interface adjustment system **900** includes a wearable device, which is the eyewear device **100** with an eye movement tracker **213** (e.g., shown as infrared emitter **215** and infrared camera **220** in FIG. 2B). User interface adjustments system **900** also includes a mobile device **990** and a server system **998** connected via various networks. Mobile device **990** may be a smartphone, tablet, laptop computer, access point, or any other such device capable of connecting with eyewear device **100** using both a low-power wireless connection **925** and a high-speed wireless connection **937**. Mobile device **990** is connected to server system **998** and network **995**. The network **995** may include any combination of wired and wireless connections.

[0098] Eyewear device **100** includes at least two visible light cameras **114A-B** (one associated with the left lateral

side 170A and one associated with the right lateral side 170B). Eyewear device 100 further includes two see-through image displays 180C-D of the optical assembly 180A-B (one associated with the left lateral side 170A and one associated with the right lateral side 170B). Eyewear device 100 also includes image display driver 942, image processor 912, low-power circuitry 920, and high-speed circuitry 930. The components shown in FIG. 9 for the eyewear device 100/200 are located on one or more circuit boards, for example a PCB or flexible PCB, in the temples 110A-B as previously described. Alternatively, or additionally, the depicted components can be located in the temples, frames, hinges, or bridge of the eyewear device 100. Left and right visible light cameras 114A-B can include digital camera elements such as a complementary metal-oxide-semiconductor (CMOS) image sensor, charge coupled device, a lens, or any other respective visible or light capturing elements that may be used to capture data, including images of scenes with unknown objects.

[0099] Eye movement tracking programming implements the user interface field of view adjustment instructions, including, to cause the eyewear device 100 to track, via the eye movement tracker 213, the eye movement of the eye of the user of the eyewear device 100. Other implemented instructions (functions) cause the eyewear device 100 to determine, a field of view adjustment to the initial field of view of an initial displayed image based on the detected eye movement of the user corresponding to a successive eye direction. Further implemented instructions generate a successive displayed image of the sequence of displayed images based on the field of view adjustment. The successive displayed image is produced as visible output to the user via the user interface. This visible output appears on the see-through image displays 180C-D of optical assembly 180A-B, which is driven by image display driver 942 to present the sequence of displayed images, including the initial displayed image with the initial field of view and the successive displayed image with the successive field of view.

[0100] As shown in FIG. 9, high-speed circuitry 930 includes high-speed processor 932, memory 934, and high-speed wireless circuitry 936. In the example, the image display driver 942 is coupled to the high-speed circuitry 930 and operated by the high-speed processor 932 in order to drive the left and right image displays 180C-D of the optical assembly 180A-B to create the virtual image. High-speed processor 932 may be any processor capable of managing high-speed communications and operation of any general computing system needed for eyewear device 100. High-speed processor 932 includes processing resources needed for managing high-speed data transfers on high-speed wireless connection 937 to a wireless local area network (WLAN) using high-speed wireless circuitry 936. In certain examples, the high-speed processor 932 executes an operating system such as a LINUX operating system or other such operating system of the eyewear device 100 and the operating system is stored in memory 934 for execution. In addition to any other responsibilities, the high-speed processor 932 executing a software architecture for the eyewear device 100 is used to manage data transfers with high-speed wireless circuitry 936. In certain examples, high-speed wireless circuitry 936 is configured to implement Institute of Electrical and Electronic Engineers (IEEE) 802.11 communication standards, also referred to herein as Wi-Fi. In other

examples, other high-speed communications standards may be implemented by high-speed wireless circuitry 936.

[0101] Low-power wireless circuitry 924 and the high-speed wireless circuitry 936 of the eyewear device 100 can include short range transceivers (e.g., UWB or Bluetooth™) and wireless wide, local, or wide area network transceivers (e.g., cellular or WiFi) including antennas 808. Mobile device 990, including the transceivers communicating via the low-power wireless connection 925 and high-speed wireless connection 937, may be implemented using details of the architecture of the eyewear device 100, as can other elements of network 995.

[0102] Memory 934 includes any storage device capable of storing various data and applications, including, among other things, color maps, camera data generated by the left and right visible light cameras 114A-B and the image processor 912, as well as images generated for display by the image display driver 942 on the see-through image displays 180C-D of the optical assembly 180A-B. While memory 934 is shown as integrated with high-speed circuitry 930, in other examples, memory 934 may be an independent stand-alone element of the eyewear device 100. In certain such examples, electrical routing lines may provide a connection through a chip that includes the high-speed processor 932 from the image processor 912 or low-power processor 922 to the memory 934. In other examples, the high-speed processor 932 may manage addressing of memory 934 such that the low-power processor 922 will boot the high-speed processor 932 any time that a read or write operation involving memory 934 is needed.

[0103] Server system 998 may be one or more computing devices as part of a service or network computing system, for example, that include a processor, a memory, and network communication interface to communicate over the network 995 with the mobile device 990 and eyewear device 100. Eyewear device 100 is connected with a host computer. For example, the eyewear device 100 is paired with the mobile device 990 via the high-speed wireless connection 937 or connected to the server system 998 via the network 995.

[0104] Output components of the eyewear device 100 include visual components, such as the left and right image displays 180C-D of optical assembly 180A-B as described in FIGS. 2C-D (e.g., a display such as a liquid crystal display (LCD), a plasma display panel (PDP), a light emitting diode (LED) display, a projector, or a waveguide). The image displays 180C-D of the optical assembly 180A-B are driven by the image display driver 942. The output components of the eyewear device 100 further include acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components of the eyewear device 100, the mobile device 990, and server system 998, may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

[0105] Eyewear device 100 may optionally include additional peripheral device elements. Such peripheral device

elements may include heat sensors **940**, biometric sensors, additional sensors, or display elements integrated with eyewear device **100**. For example, peripheral device elements **919** may include any I/O components including output components, motion components, position components, or any other such elements described herein. The eyewear device **100** can take other forms and may incorporate other types of frameworks, for example, a headgear, a headset, or a helmet.

[0106] For example, the biometric components of the user interface field of view adjustment **900** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The position components include location sensor components to generate location coordinates (e.g., a Global Positioning System (GPS) receiver component), WiFi or Bluetooth™ transceivers to generate positioning system coordinates, altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like. Such positioning system coordinates can also be received over wireless connections **925** and **937** from the mobile device **990** via the low-power wireless circuitry **924** or high-speed wireless circuitry **936**.

[0107] According to some examples, an “application” or “applications” are program(s) that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, a third party application (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating systems. In this example, the third-party application can invoke API calls provided by the operating system to facilitate functionality described herein.

[0108] FIG. 10 is a flowchart **1000** illustrating the operation of the slot antenna **890**. These blocks do not need to be performed sequentially, and the processing and wireless communications via antennas **880A** and **880B** can be performed simultaneously.

[0109] At block **1002**, the slot antenna **890** shown in FIG. 8L is energized with an RF signal at frequency f_1 . This is done by feeding the RF signal to the slot antenna **880A** via the antenna feed **882A** and the slot antenna **880B** via antenna feed **882B**.

[0110] At block **1004**, the inner ring **844** of antenna **880A** and the outer conductive material **804** forming the outer ring radiate an E-field at the frequency f_1 in a first direction d_1 . This E-field is shown in FIG. 8L.

[0111] At block **1006**, the inner ring **844** of antenna **880B** and the outer conductive material **804** forming the outer ring

radiate an E-field in a second direction d_2 that is orthogonal to the first direction d_1 . This is also shown in FIG. 8L.

[0112] The slot antenna **890** having two slot-ring antennas **880A** and **880B** co-exist in very close proximity to each other and support the same frequency band (in this example f_1), with minimal coupling to each other because of the orthogonal E-fields. The slot antenna **890** is integrated into eyewear **100** such that the eyewear supports wireless communications, and it has increased E-fields that are orthogonal to each other.

[0113] It will be understood that the terms and expressions used herein have the ordinary meaning as is accorded to such terms and expressions with respect to their corresponding respective areas of inquiry and study except where specific meanings have otherwise been set forth herein. Relational terms such as first and second and the like may be used solely to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “includes,” “including,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises or includes a list of elements or steps does not include only those elements or steps but may include other elements or steps not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “a” or “an” does not, without further constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

[0114] Unless otherwise stated, any and all measurements, values, ratings, positions, magnitudes, sizes, and other specifications that are set forth in this specification, including in the claims that follow, are approximate, not exact. Such amounts are intended to have a reasonable range that is consistent with the functions to which they relate and with what is customary in the art to which they pertain. For example, unless expressly stated otherwise, a parameter value or the like may vary by as much as $\pm 10\%$ from the stated amount.

[0115] In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, the subject matter to be protected lies in less than all features of any single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

[0116] While the foregoing has described what are considered to be the best mode and other examples, it is understood that various modifications may be made therein and that the subject matter disclosed herein may be implemented in various forms and examples, and that they may be applied in numerous applications, only some of which have been described herein. It is intended by the following claims to claim any and all modifications and variations that fall within the true scope of the present concepts.

What is claimed is:

1. An eyewear, comprising:
 - a frame;
 - an optical assembly supported by the frame; and

an antenna coupled to the frame and configured to communicate wireless signals at a frequency, wherein the antenna comprises a first slot antenna and a second slot antenna each configured to generate a respective radiating E-field in a direction that is orthogonal to the other, wherein the first slot antenna has a first inner ring and a first outer ring encompassing the first inner ring, and the second slot antenna has a second inner ring and a second outer ring encompassing the second inner ring, wherein the first slot antenna and the second slot antenna each have a respective antenna feed and a ring-short configured to create the respective radiating E-fields that are orthogonal to each other, and wherein a segment is interposed between the first outer ring and the second outer ring.

2. The eyewear of claim 1, wherein the segment has a width that is less than a width of the first outer ring and the second outer ring.

3. The eyewear of claim 2, wherein the first outer ring, the second outer ring and the segment form a ground plane.

4. The eyewear of claim 1, wherein the frame has a first side and an opposing second side with a bridge extending between the first side and the second side, wherein the segment is disposed in the bridge.

5. The eyewear of claim 4, wherein the optical assembly comprises a first see-through display supported by the frame first side and a second see-through assembly supported by the frame second side.

6. The eyewear of claim 1, wherein the antenna feed of the first slot antenna is offset 90 degrees with respect to the segment and the antenna feed of the second slot antenna is offset 180 degrees from the segment.

7. The eyewear of claim 1, wherein the ring-short of the first slot antenna is disposed proximate the segment and the ring-short of the second slot antenna is offset 90 degrees with respect to the segment.

8. The eyewear of claim 7, wherein the ring-short of the first slot antenna is disposed proximate the segment and the ring-short of the second slot antenna is offset 90 degrees with respect to the segment.

9. The eyewear of claim 1, wherein the antenna feed of the first slot antenna is offset 90 degrees with respect to the ring-short of the first slot antenna.

10. The eyewear of claim 1, further comprising an inductor-capacitor (LC) circuit coupled between the first inner ring and the first outer ring.

11. A method of using eyewear, the eyewear having a frame, an optical assembly supported by the frame, and an antenna coupled to the frame and configured to communi-

cate wireless signals at a frequency, wherein the antenna comprises a first slot antenna and a second slot antenna each configured to generate a respective radiating E-field in a direction that is orthogonal to the other, wherein the first slot antenna has a first inner ring and a first outer ring encompassing the first inner ring, and the second slot antenna has a second inner ring and a second outer ring encompassing the second inner ring, wherein the first slot antenna and the second slot antenna each have a respective antenna feed and a ring-short configured to create the respective radiating E-fields that are orthogonal to each other, and wherein a segment is interposed between the first outer ring and the second outer ring, the method comprising:

the first slot antenna radiating a first E-field in a first direction; and

the second slot antenna radiating a second E-field in a second direction that is orthogonal to the first direction.

12. The method of claim 11, wherein the segment has a width that is less than a width of the first outer ring and the second outer ring.

13. The method of claim 12, wherein the first outer ring, the second outer ring and the segment form a ground plane.

14. The method of claim 11, wherein the frame has a first side and an opposing second side with a bridge extending between the first side and the second side, wherein the segment is disposed in the bridge.

15. The method of claim 14, wherein the optical assembly comprises a first see-through display supported by the frame first side and a second see-through assembly supported by the frame second side.

16. The method of claim 11, wherein the antenna feed of the first slot antenna is offset 90 degrees with respect to the segment and the antenna feed of the second slot antenna is offset 180 degrees from the segment.

17. The method of claim 11, wherein the ring-short of the first slot antenna is disposed proximate the segment and the ring-short of the second slot antenna is offset 90 degrees with respect to the segment.

18. The method of claim 11, wherein the ring-short of the first slot antenna is disposed proximate the segment and the ring-short of the second slot antenna is offset 90 degrees with respect to the segment.

19. The method of claim 11, wherein the antenna feed of the first slot antenna is offset 90 degrees with respect to the ring-short of the first slot antenna.

20. The method of claim 11, further comprising an inductor-capacitor (LC) circuit coupled between the first inner ring and the first outer ring.

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